

The Economics of the Global Water Cycle

An Integrated Assessment Model

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Introduction

Economic & pop. growth → pressure on water resources (Rodell et al., 2018, Nature)

- Globalization → water-intensive specialization (agriculture) in Brazil, China, India, US, ... (Debaere, 2014; Copeland et al., 2022; Carleton et al., 2024b, 2025)
- Concerning water use (w.r. to recharge) → billions exposed to future water insecurity (Mekonnen and Hoekstra, 2016, 2020; Greaney et al., 2026, in e.g., Nature & Science)

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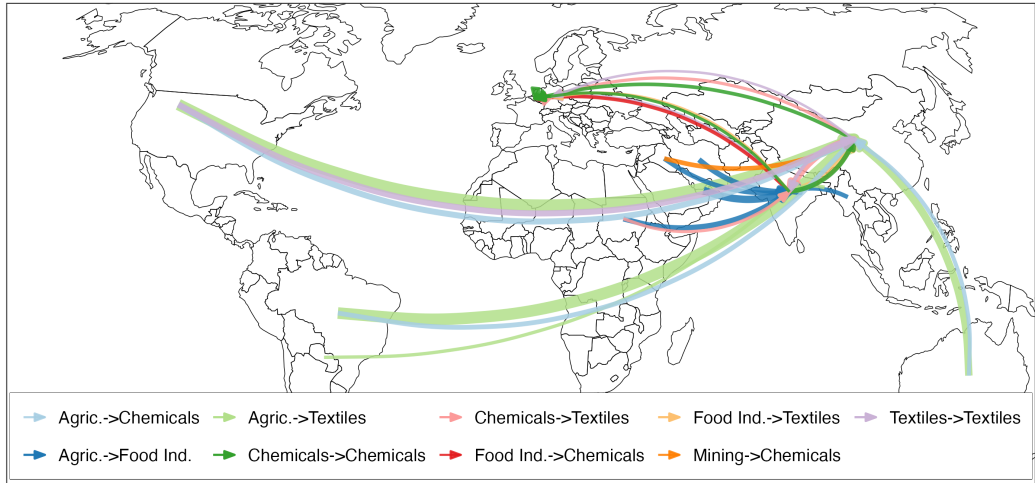
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This paper – these **local challenges reflect a global problem** due to spatial linkages

- Traded goods embody water footprints
- Water resources are linked through the global water cycle

Globalized economy → globalized water footprint...

Fig. Water footprint (water m³/USD) of Germany's textile industry (GLORIA data, 2020)



... amplified by the global water cycle

Fig. Evaporation-precipitation flows from the Mississippi Basin (RECON data; De Petrillo et al., 2025a,b)



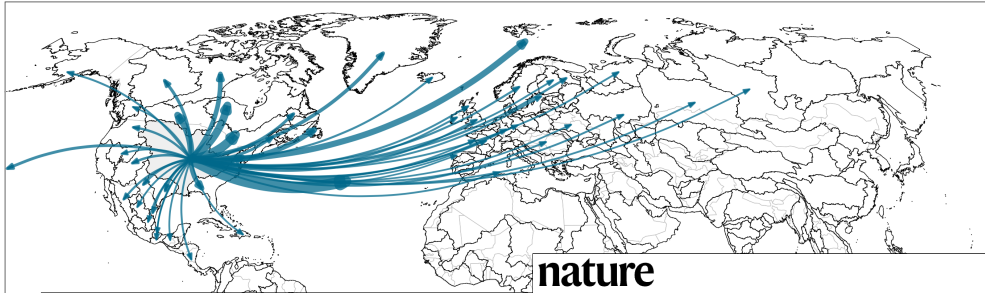
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nature

**Why we need a new economic
water as a common good**

Anthropogenic pressures and climate change are altering water flows worldwide. Better understanding, new economic thinking and an international governance framework are needed to stave off catastrophe.

nature

**Carbon's social cost can't be
retrofitted to water**

What we do

How does the co-evolution of economy & water cycle → water security and welfare?

- Can policies correct all (spatial and intertemporal) unpriced externalities?*

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1. Empirically show that the economy and water cycle are highly interconnected
 - Climate-economy interaction → anthropogenic distortion of the water cycle
2. Develop a dynamic spatial model integrated with the global water cycle
 - Common pool resource externalities → **globalized tragedy of the commons**

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2. Develop a dynamic spatial model integrated with the global water cycle
 - Common pool resource externalities → **globalized tragedy of the commons**
3. Derive an optimal water policy rule to internalize all unpriced externalities
 - Pigouvian price on water ↔ spatial "centrality" of water resources

Related literature

policy contribution

Spatial-climate economics (e.g., Nordhaus, 1994; Desmet and Rossi-Hansberg, 2024) [more](#)

- Study the spatial climate-economy link + the role of trade, taxes, and policies
- Contribution: *the first integrated assessment model of the global water cycle*

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Hydrology vs. water economics (e.g., Araujo, 2023; Niazi et al., 2024; Carleton et al., 2025)

- Rich models that yet lack of full integration (or context specific) [water economics](#)
- Contribution: *bridge both fields, integrated framework for policy evaluation*

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Resource econ. (e.g., Hotelling, 1931; Vanek, 1959; Solow, 1974; Davis and Weinstein, 2001)

- On the "limits to growth" & resource content of trade more
- *Study the optimal exploitation of spatially interconnected resources*

Roadmap

1. Data & motivating facts
2. A dynamic spatial model of the global economy & water cycle
 - Integrated model and stylised simulations
 - Optimal water policy for an interconnected world
3. Current work: linking theory to data + counterfactuals + ...?

**Motivating facts: how interconnected
are the global water cycle & economy?**

Preamble: a closed-system, spatially connected water cycle

Without economic water use, water stocks R in (any) geography i, j evolves as:

$$R_i(t+1) = R_i(t) - E_i(t) - S_i(t) + \underbrace{\tilde{E}_i(t)}_{\sum_j f_{ji}^{(E)} E_j(t)} + \underbrace{\tilde{S}_i(t)}_{\sum_j f_{ji}^{(S)} S_j(t)}$$

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↑
Outflows (evaporation & surface runoffs)

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Atmospheric linkages

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Economic water use $H_i(t)$ distorts eq. (1) – but in an ambiguous direction

- **Closed global system** $\rightarrow \sum_i R_i(t) = \sum_i R_i(t+1) \forall t$
- Withdraws **return to water system** (e.g., evapotranspiration, surface runoff)

Next: empirical facts that corroborate these mechanisms (e.g., strength of $\{f_{ji}^{(E)}\}_{j,i}$)

Fact 1: global atmospheric links between water resources

RECON data: **atmospheric moisture flows** (0.5°

$\times 0.5^\circ$, De Petrillo et al., 2025a,b) [details](#)

- Evap. $\rightarrow$ Precip. flows (m^3 , $\sim 2008-17$)

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Our focus: major land water basins (GAEZ)

- Basins $\equiv$ precipitation catchment areas (no surface runoff b/w basins)

Fig. Land Water Basins from FAO GAEZ



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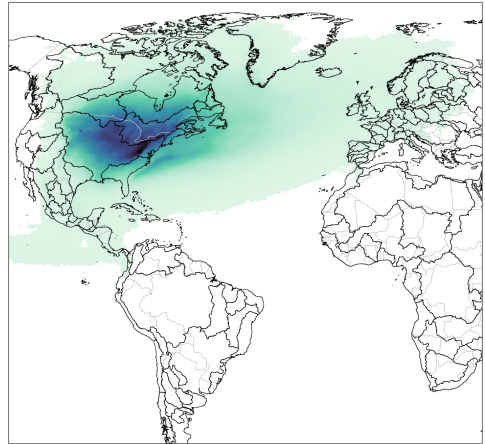
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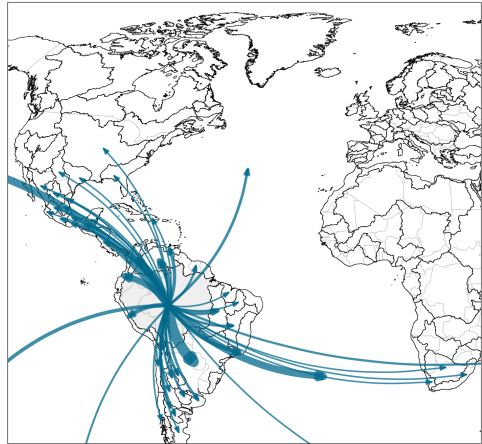
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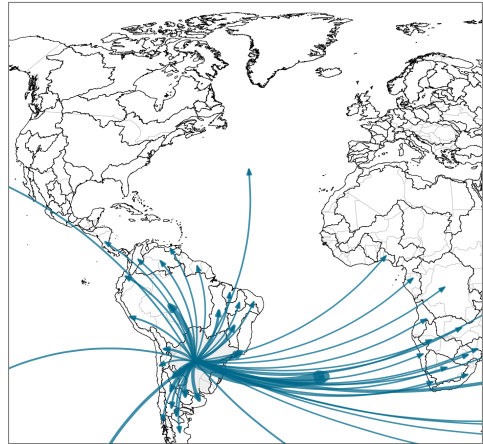
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Table 1: Water "import shares" per origin type (%) [maps](#) [exports](#) [surface flows](#)

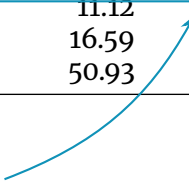
	Own basin	Other basins (domestic)	Other basins (foreign)	Oceans
Min.	0.65	0.09	0.17	5.82
Q1	2.41	3.68	7.94	52.99
Median	4.81	8.62	17.71	68.72
Mean	7.68	11.12	20.04	63.99
Q3	9.61	16.59	28.58	79.09
Max.	48.66	50.93	71.27	98.18

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Median basin ~ 25% from other land basins



Fact 2: atmospheric & surface flows are equally important

How large are atmospheric flows vis-à-vis (country-level) total recharge $\tilde{E}_i + \tilde{S}_i$? ●

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Table 2: Country-level water "import shares" per origin type (%)

	Ocean	Own	Foreign atm.	Foreign surf.	Foreign total
Min.	1.12	0.02	0.02	0.00	0.02
Q1	34.67	0.90	8.33	0.00	16.47
Median	56.34	2.53	20.53	3.99	35.23
Mean	56.65	5.84	22.47	15.04	37.51
Q3	79.13	7.41	33.93	19.24	55.83
Max.	99.80	37.87	74.47	98.52	98.80

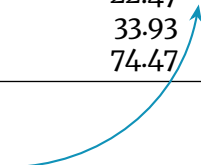
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Foreign precipitation $\sim 20\%$ total recharge
(and the largest foreign recharge share)



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Does the global economy **internalize these spatial externalities?** **Apparently no.**

- Most water-stressed countries $\leftrightarrow$ top "atmospheric water exporters" [details](#)

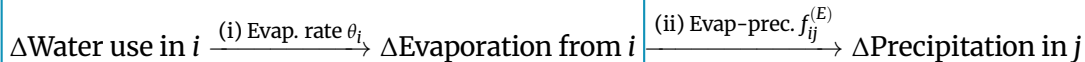
Fact 3: local water use propagates via the global water cycle

Does **economic water use** $\leftrightarrow$ observed Δ water cycle? Ideal (long-term) experiment:

Δ Water use in $i \xrightarrow{\text{(i) Evap. rate } \theta_i} \Delta$ Evaporation from $i \xrightarrow{\text{(ii) Evap-prec. } f_{ij}^{(E)}} \Delta$ Precipitation in j

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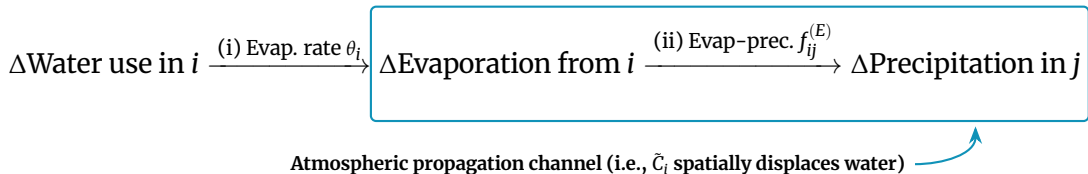
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Consumptive water use channel (i.e., $\tilde{C}_i \equiv \text{Water use}_i \times \theta_i$)

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$$\Delta \text{Precipitation}_j = \alpha \sum_i f_{ij}^{(E)} \Delta \tilde{C}_i + \beta \Delta X_j + \gamma_{FE} + \varepsilon_j, \quad (2)$$

where $\Delta \equiv 1960\text{--}2015$ long diff. (Burke and Emerick, 2016; Carleton et al., 2024a) [raw data](#)

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Observed long-term Δ Precipitation
(GPCC data; Gauge filter ≥ 1) [details](#)

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Exposure to (global) consumptive water use (i.e., $\tilde{C}_i \forall i$)

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- Δ Consumptive water use $\tilde{C}_i \sim \Delta$ grid cell-level irrigation share (in SD) [data details](#)
- $\gamma_{FE} \equiv \text{region} \times \text{aridity quintile} \times \text{latitude band} + \text{Bartik IV for } \Delta \tilde{C}_i$ [IV details](#)

Fact 3: local water use propagates via the global water cycle

Table 3: Estimation Results (OLS and IV)

	Δ Precipitation			
	(1)	(2)	(3)	(4)
Δ Irrigation Exposure	0.0331* (0.0181)	0.0353* (0.0198)	0.2133*** (0.0718)	0.2529*** (0.0759)
Bartik IV	No	No	Yes	Yes
Weights	None	Cropland	None	Cropland
Observations	5,367	5,031	5,367	5,031
Kleibergen-Paap F-Stat			22.434	31.296

Conley (500km) standard-errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.
All regression include climate controls (ΔX_j) and Region-Aridity quintile-Latitude FE (α_{FE}).

Magnitude interpretation (3): $\hat{\alpha} = 0.21$ per SD & $SD(\Delta \text{ irrig. exposure}) \sim 0.01 \rightarrow$
– Moving from P10 to P90 (~ 1.3 SD) $\sim$ **33% more precipitation** ($\sim 0.6\%$ /year)

sum. stats

1st stage

no controls

alternative IV

irrig. intensity

annual precip.

all cells

imported vs. domestic

placebo

horse race

Motivating facts: taking stock

Global economy & water cycle – highly interconnected networks

- Local water exploitation → sizeable global consequences (via feedback loops)
- Current water challenges: cross-sectional evidence of a **slow-moving process**

How will the economy & water co-evolve, and where it will lead is unknown

Next: an integrated model that incorporates such interconnections

- Optimal water policy that corrects all water-related externalities

**Theory: a dynamic spatial model of
the global economy & water cycle**

Dynamic spatial model – overview

Economic module: trade model with labor and water as inputs

- Subject to policies: tariffs, subsidies, water prices
- Water use → higher extraction/prod. costs → shifts in trade & production

Water cycle module: closed system of geographical locations whose stock are

- Depleted by evaporation, runoffs, & water use; replenished by inflows
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Water-Economy $\hookrightarrow$ Feedback → Globalized Tragedy of the Commons

(i.e., local extraction imposes unpriced costs on the rest of the world)

Dynamic spatial model – environment & preferences

Geography of N locations i, j ; fixed labor $\bar{L}_i$ and endogenous water stocks $R_i(t)$

- $K + 1$ industries ($k = 0$: water extraction; $k \geq 1$: final goods), time $t = 0, 1, \dots$

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Policy instruments create a wedge b/w producer (P) and consumer prices ($\tilde{P}$):

- Import tariffs $t_{ji,k}$, export & production subsidies $x_{ij,k}$, $s_{i,k}$, and **water pricing** $\tau_{i,k}$
- Tax revenues T_i rebated to households $\rightarrow$ balanced budget: $I_i = w_i \bar{L}_i + T_i$

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Demand: $W_i = V_i(I_i, \tilde{P}_i)$; Marshallian demand (Roy's identity) $C_{ji,k} = D_{ji,k}(I_i, \tilde{P}_i)$

- Versatile enough for any demand system (e.g., CD-CES, non-homothetic CES)

Dynamic spatial model – technology & water extraction

Supply: Final goods ($k \geq 1$) with CRS production $(L_{j,k}, H_{j,k})$; producer price:

$$P_{ji,k} = \bar{d}_{ji,k} c_{j,k}(w_j, \tilde{r}_{j,k}), \quad \text{where}$$

$\tilde{r}_{j,k} = r_j + \tau_{j,k} \equiv$ effective water price (inclusive of **water pricing** $\tau_{j,k}$)

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Water extraction ($k = 0$) uses labor as $Q_{j,0} = L_{j,0} / G_j(R_j) \rightarrow$

$$r_j = w_j G_j(R_j)$$

where $G'_j < 0 \rightarrow$ extraction costs increase as resources deplete (and $H_{j,k}$ decreases)

Dynamic spatial model – technology & water extraction

Supply: Final goods ($k \geq 1$) with CRS production $(L_{j,k}, H_{j,k})$; producer price:

$$P_{ji,k} = \bar{d}_{ji,k} c_{j,k}(w_j, \tilde{r}_{j,k}), \quad \text{where}$$

$\tilde{r}_{j,k} = r_j + \tau_{j,k} \equiv$ effective water price (inclusive of **water pricing** $\tau_{j,k}$)

Water extraction ($k = 0$) uses labor as $Q_{j,0} = L_{j,0} / G_j(R_j) \rightarrow$

$$r_j = w_j G_j(R_j)$$

where $G'_j < 0 \rightarrow$ extraction costs increase as resources deplete (and $H_{j,k}$ decreases)

Myopic agents $\rightarrow$ **Common Pool Resource Externality** (spatially and intertemporally)

(i.e., firms take r_j as given, not internalizing that withdrawals affect $R_j \forall j$)

Water cycle – global geography & natural hydrology

Water cycle: **closed system** of land geographies $\mathcal{B} = \{1, \dots, B\}$ + ocean $\mathcal{O} = \{0\}$

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \sum_j f_{ji}^{(E)} \kappa_j R_j(t) + \sum_j f_{ji}^{(S)} \delta_j R_j(t)$$

Water cycle – global geography & natural hydrology

Water cycle: **closed system** of land geographies $\mathcal{B} = \{1, \dots, B\}$ + ocean $\mathcal{O} = \{0\}$

$$R_i(t+1) = R_i(t) - \underbrace{E_i(t)}_{\kappa_i R_i(t)} - \underbrace{S_i(t)}_{\delta_i R_i(t)} + \sum_j f_{ji}^{(E)} \kappa_j R_j(t) + \sum_j f_{ji}^{(S)} \delta_j R_j(t)$$

Oceans as exogenous reservoirs (i.e., $E_0 = \bar{E}_0$, $S_0 = \bar{S}_0$) $\rightarrow$ **law of motion** $i \geq 1$:

$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \mathbf{\Phi} \mathbf{R}_B(t), \text{ where}$$

- $\mathbf{\Gamma} \equiv \mathbf{F}_{OB}^{(E)} \bar{\mathbf{E}}_0 \equiv$ *exogenous ocean precipitation inflows (oceans $\sim$ "sink" $\rightarrow \bar{S}_0 = 0$)*
- $\mathbf{\Phi} \equiv \mathbf{I} - \mathbf{\kappa} - \mathbf{\delta} + \mathbf{F}_{BB}^{(E)} \mathbf{\kappa} + \mathbf{F}_{BB}^{(S)} \mathbf{\delta} \equiv$ *net retention (losses + atm. & surface recycling)*

Water cycle – economic withdrawals & consumptive use

Now **add economic activity** $H_{ik}(t)$ with consumptive-use share $\theta_{ik} \in (0, 1)$:

- Fraction of $H_{ik}(t)$ lost via evaporation; remainder returns as runoff

Water cycle – economic withdrawals & consumptive use

Now **add economic activity** $H_{ik}(t)$ with consumptive-use share $\theta_{ik} \in (0, 1)$:

- Fraction of $H_{ik}(t)$ lost via evaporation; remainder returns as runoff

Location-level aggregates:

$$H_i(t) = \sum_k H_{ik}(t), \quad \theta_i(t) = \frac{\sum_k \theta_{ik} H_{ik}(t)}{\sum_k H_{ik}(t)} \quad (\text{time-varying with sectoral mix})$$

Modified evaporation (natural + economic):

$$E_B(t) = \kappa R_B(t) + \theta(t) H(t)$$

Economic withdrawals $\uparrow$ evaporation, shifting precipitation patterns worldwide

Water cycle – updated law of motion, steady state, & gen. eq.

Law of motion with economic activity:

$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \mathbf{\Phi} \mathbf{R}_B(t) + \mathbf{\Psi}(t) \mathbf{H}(t), \quad \mathbf{\Psi}(t) \equiv (\mathbf{F}_{BB}^{(E)} - \mathbf{I}) \boldsymbol{\theta}(t)$$

$\mathbf{\Psi}(t) \equiv$ *net economic retention* $\rightarrow \mathbf{\Psi}(t) \mathbf{H}(t)$ has an ambiguous sign

Water cycle – updated law of motion, steady state, & gen. eq.

Law of motion with economic activity:

$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \mathbf{\Phi} \mathbf{R}_B(t) + \mathbf{\Psi}(t) \mathbf{H}(t), \quad \mathbf{\Psi}(t) \equiv (\mathbf{F}_{BB}^{(E)} - \mathbf{I}) \boldsymbol{\theta}(t)$$

$\mathbf{\Psi}(t) \equiv$ *net economic retention* $\rightarrow \mathbf{\Psi}(t) \mathbf{H}(t)$ has an ambiguous sign

Steady state with economic activity (fixed-point in $\mathbf{R}_B^{(ss)}$):

$$\mathbf{R}_B^{(ss)} = (\mathbf{I} - \mathbf{\Phi})^{-1} \left(\mathbf{\Gamma} + \mathbf{\Psi}^{(ss)} \mathbf{H}(\mathbf{R}_B^{(ss)}) \right)$$

Closed system: $\sum_i R_i(t+1) = \sum_i R_i(t)$ – fixed global water stock

Water cycle – updated law of motion, steady state, & gen. eq.

Law of motion with economic activity:

$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \mathbf{\Phi} \mathbf{R}_B(t) + \mathbf{\Psi}(t) \mathbf{H}(t), \quad \mathbf{\Psi}(t) \equiv (\mathbf{F}_{BB}^{(E)} - \mathbf{I}) \boldsymbol{\theta}(t)$$

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Closed system: $\sum_i R_i(t+1) = \sum_i R_i(t)$ – fixed global water stock

General equilibrium: $\{w_i(t)\}_{i,t}$ satisfying LOM & $\{w_i^{(ss)}\}_i$ satisfying SS above [details](#)

Optimal water policy for an interconnected world

Optimal water policy – problem & Pigouvian tax

[derivation](#)[illustration](#)

Optimal policy $\mathbb{I}_i \equiv \{\mathbf{t}_i, \mathbf{x}_i, \mathbf{s}_i, \tau_i\}$ + transfers Δ_i ($\sum_i \Delta_i = 0$) solves

$$\max_{\mathbb{I}} \mathcal{W}(W_1(\mathbb{I}), \dots, W_N(\mathbb{I})) \quad \text{s.to steady-state GE relationships}$$

Optimal water policy – problem & Pigouvian tax

[derivation](#)[illustration](#)

Optimal policy $\mathbb{I}_i \equiv \{t_i, x_i, s_i, \tau_i\}$ + transfers Δ_i ($\sum_i \Delta_i = 0$) solves

$$\max_{\mathbb{I}} \mathcal{W}(W_1(\mathbb{I}), \dots, W_N(\mathbb{I})) \quad \text{s.to steady-state GE relationships}$$

Non-water taxes: $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$ (distorting goods prices is inefficient)

Optimal water policy – problem & Pigouvian tax derivation illustration

Optimal policy $\mathbb{I}_i \equiv \{\mathbf{t}_i, \mathbf{x}_i, \mathbf{s}_i, \tau_i\}$ + transfers Δ_i ($\sum_i \Delta_i = 0$) solves

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Non-water taxes: $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$ (distorting goods prices is inefficient)

Optimal water tax (Pigouvian + spatial linkages):

$$\tau_{i,k} = \tilde{r}_{i,k} - r_i = \sum_n \frac{G'_n(\cdot)}{G_n(\cdot)} r_n \sum_g \lambda_{in,g} H_{n,g}$$

- $G'_n/G_n < 0$: *extraction externality* – depletion $\rightarrow$ higher future unit costs
- $\lambda_{in,g}$: *spatial weights* – hydrological links propagate the externality globally

$\tau_{i,k} \equiv$ Pigouvian price $\sim$ **location-specific Social Cost of Water** ($\Delta_i \leftrightarrow$ distr. goals)

**Quantitative exercise: real-world calibration,
simulations, and the social cost of water**

Quantitative exercise – overview (in progress)

1. Calibrate the model with global economic (2025 ca.) and hydrological data

- (i) **Hydrological parameters:** $\kappa_i, \delta_i, \Gamma_i, f_{ij}^{(E)}, f_{ij}^{(S)}$ (✓) ●
- (ii) **Water use parameters:** θ_{ik} (consumptive-use shares) (✓), extraction G_i ●

2. Status-quo simulation of the world economy & water cycle (2025–2100)

- Exogenous inputs: population and TFP growth (until 2100); steady state afterwards ●

3. Estimate the Social Cost of Water for the (steady state) global economy

- Benchmark it against the inefficient SS & alternative policies (e.g., trade liberalization) ●

Final remarks & next steps

Final remarks & next steps

Economic activity distorts the global water cycle through spatial linkages

- Slow-moving but empirically measurable process

A dynamic spatial model integrating both networks

- Common-pool externalities → *globalized tragedy of the commons*
- Corrected by a **location-specific Pigouvian water tax** ~ **Social Cost of Water**

Next steps: link to global data → quantify welfare costs of water externalities

- Simulate optimal policies for the 21st century
- Model extension (e.g., full-fledged dynamics, non-homothetic preferences, ...)

Thank you

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Appendix

Policy contribution: details [back](#)

Policy-relevant: addresses calls for global water governance ([Mazzucato et al., 2024](#)) and HORIZON EU's environmental research priorities

Envisioned legacy: inspire new research on

- **Social cost of water:** challenges the paradigm of an inherently local SCW ([Olmstead, 2010](#)) → first calculation of shadow prices internalizing **full spatial & intertemporal externalities** of local water use
- **Political economy** of shared (water) resources
- Water and the **green transition:** energy-intensive technologies, AI, ...

Spatial-climate economics: spatial economy–climate (Nordhaus, 1994; Hassler and Krusell, 2012; Krusell and Smith, 2022; Desmet and Rossi-Hansberg, 2015, 2024)

- Trade (Conte et al., 2021), green subsidies (Cruz and Rossi-Hansberg, 2024), CO₂ taxes (Conte et al., 2025), policy coordination (Farrokhi et al., 2025a)
- Distributional effects (Bourany, 2025a,b; Bourany and Rosenthal-Kay, 2025)

Gap: no study has focused on water as an outcome of the economy–climate system

- *This paper:* first integrated model of the global economy & global water cycle

Water economics literature [back](#)

Hydrology: rich models projecting future water challenges, but limited flexibility for policy exercises (e.g., Niazi et al., 2024)

Water economics: mostly context-specific settings

- India (e.g., Fishman et al., 2015; Sekhri, 2014; Blakeslee et al., 2023); Pakistan (Haseeb, 2024); US (e.g., Hornbeck and Keskin, 2014; Keiser and Shapiro, 2019a,b; Debaere and Li, 2020)

Gap: lack of full integration (e.g., exogenous recharge and/or context specific; Araujo et al., 2023; Araujo, 2023; Carleton et al., 2025; Araujo et al., 2026; Greaney et al., 2026)

- *This paper:* global, integrated & transparent framework **bridging both fields**

“Limits to growth”: intertemporal resource depletion (Solow, 1974; Stiglitz, 1974; Pindyck, 1978); resource content of trade (Vanek, 1959; Davis and Weinstein, 2001; Debaere, 2014)

Modern revival – sustainable exploitation in spatial models:

- Forests (Farrokhi et al., 2025b), fisheries (Kang, 2024), oil (Farrokhi, 2020; Hassler et al., 2022), and minerals (Domínguez-Iino et al., 2026)

Gap: the modern revival has not yet extended to water

- *This paper*: intertemporal exploitation of **spatially interconnected** water resources

RECON (De Petrillo et al., 2025a,b): bilateral atmospheric moisture flows between every pair of 0.5° grid cells worldwide

- **Methodology:** UTrack Lagrangian moisture tracking $\rightarrow$ Iterative Proportional Fitting (IPF) reconciliation with ERA5 reanalysis
- **Coverage:** global, $0.5^\circ \times 0.5^\circ$; average year based on 2008–2017
- **Content:** $f_{gg'}^{(E)}$ (m^3) = atmospheric moisture evaporated at cell g and precipitated at cell g'

RECON data: our application [back](#)

Basin-level aggregation $\rightarrow$ moisture matrix $F_{BB}^{(E)}$ used in the model's law of motion:

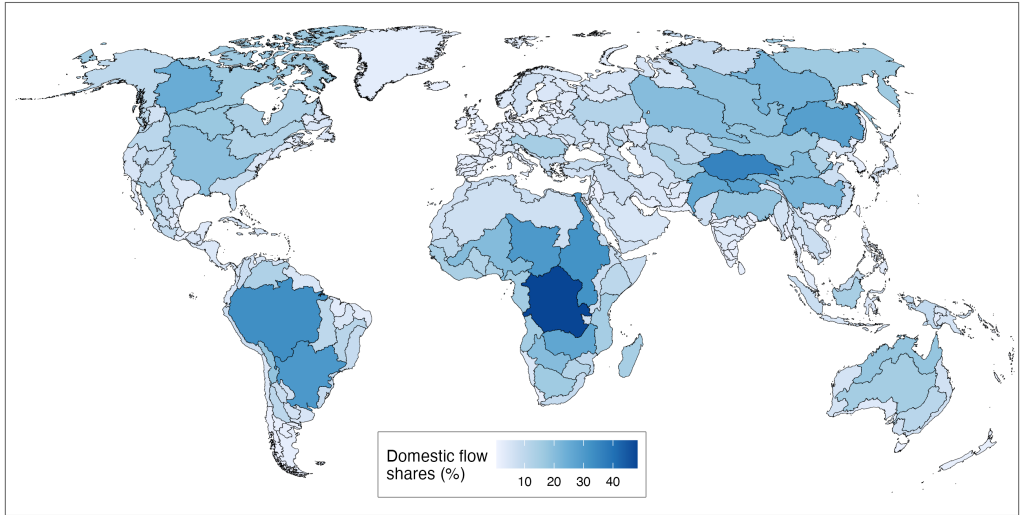
$$f_{ji}^{(E)} = \sum_{g \in j} \sum_{g' \in i} f_{gg'}^{(E)}$$

Country-level aggregation $\rightarrow$ “foreign water use” regressor in Fact 2:

$$f_{c'c}^{(E)} = \sum_{g \in c'} \sum_{g' \in c} f_{gg'}^{(E)}, \quad \Delta \text{Foreign water use}_c = \sum_{c' \neq c} f_{c'c}^{(E)} \Delta \text{Water use}_{c'}$$

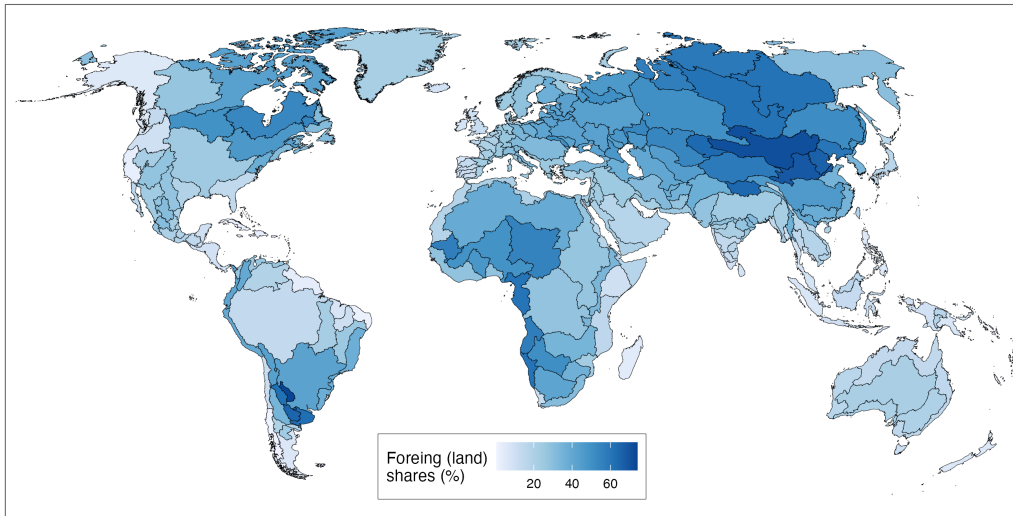
Atmospheric moisture shares: maps [back](#)

Fig. Share of atmospheric "domestic" flows



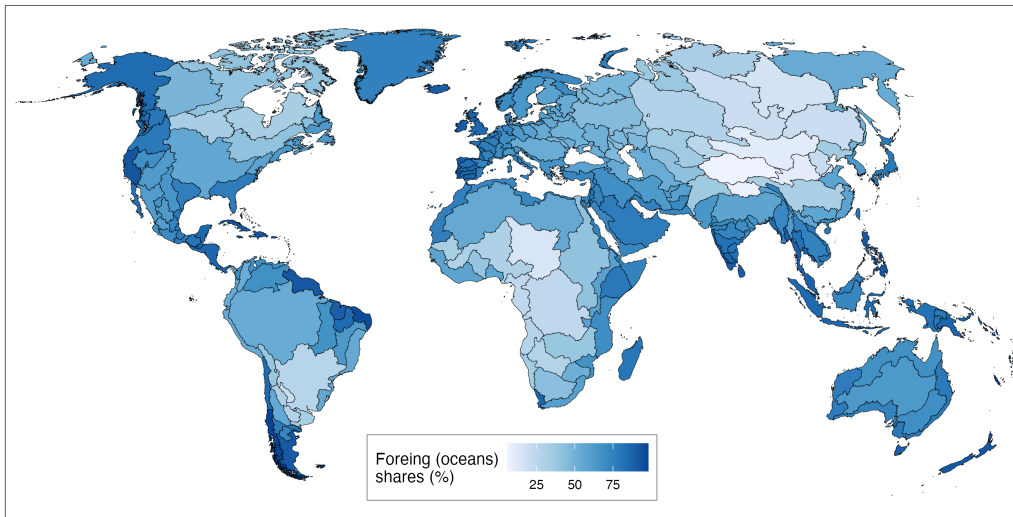
Atmospheric moisture shares: maps [back](#)

Fig. Share of atmospheric "foreign flows" from other land basins



Atmospheric moisture shares: maps [back](#)

Fig. Share of atmospheric "foreign flows" from the oceans

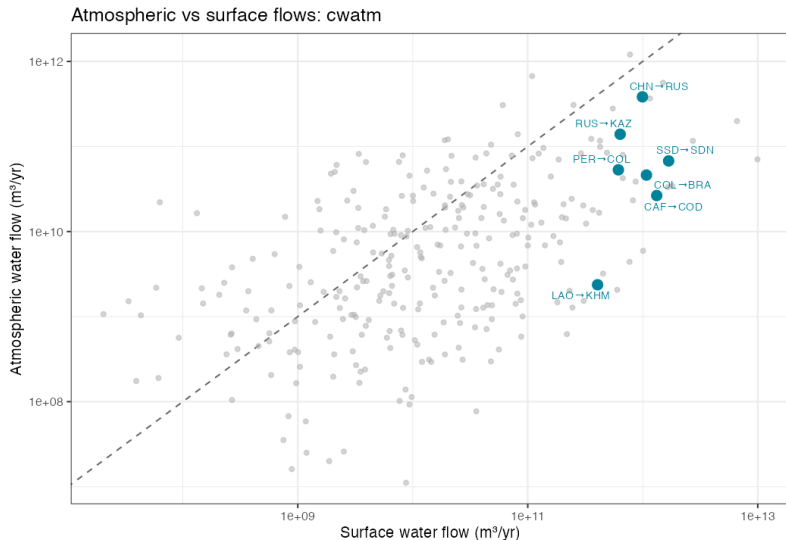


Atmospheric moisture: export shares [back](#)

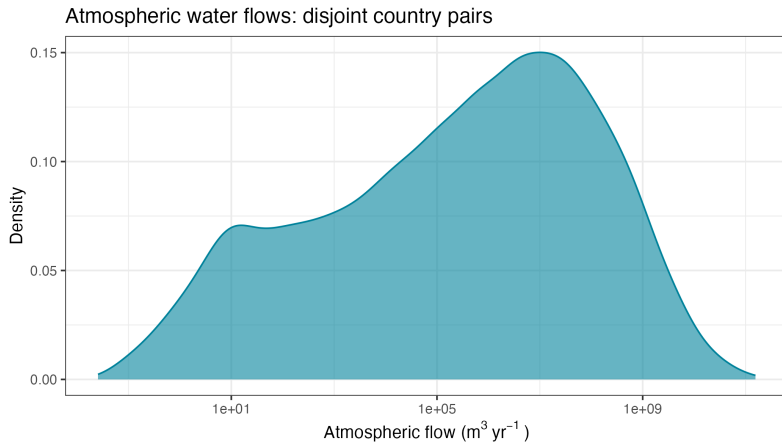
Table 4: Water "export shares" per destination basin type (%)

	Own basin	Other basins (domestic)	Other basins (foreign)	Oceans
Min.	0.77	0.06	1.39	1.60
Q1	3.01	5.59	13.63	19.43
Median	6.59	13.34	28.14	34.34
Mean	11.04	20.37	36.20	37.58
Q3	13.04	32.62	57.22	53.91
Max.	74.58	80.21	91.43	94.26

Fact 1: global atmospheric links between water resources

[back](#)

Fact 1: global atmospheric links between water resources

[back](#)

Fact 1 (bonus): top water-stressed countries $\leftrightarrow$ top "atmospheric exporters"

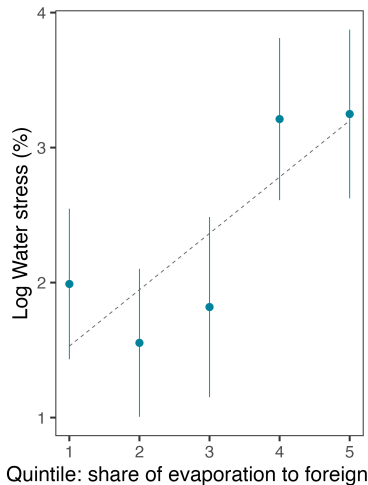
[back](#)

Water stress $\equiv$ water withdraws/endowments [FAO data](#)

Top water-stressed countries:

- Also the top "atmospheric water exporters"
- I.e., most central countries in terms of "propagation" of local water depletion

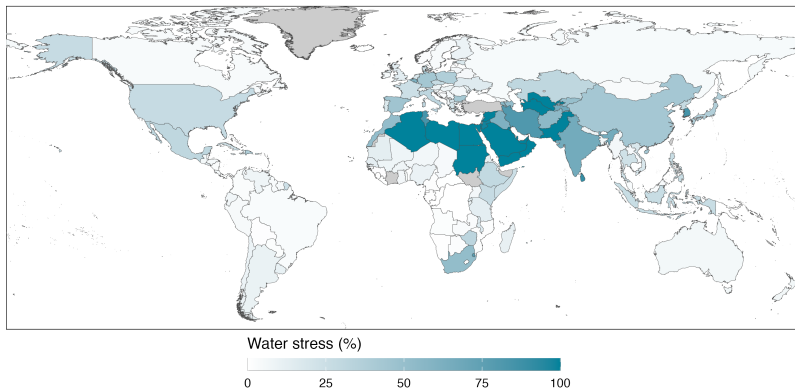
Fig. Water exporters vs. water stress



Water stress data from FAO's AQUASTAT [back](#)

Definition (FAO AQUASTAT): Water stress = $\frac{\text{Total freshwater withdrawn}}{\text{Total renewable freshwater resources}} \times 100$

Fig. Water stress (%) across countries (FAO AQUASTAT, avg. 2008–2017)



Note: Values above 100% winsorized. Grey = missing data.

Fact 2: data construction and modelling choices [back](#)

Outcome:

- $\Delta \log$ **growing-season** precipitation — GPCC [1960](#) [2015](#) [LD map](#) [gauges](#)
- NH (Apr–Sep), SH (Oct–Mar)

Endogenous variable:

- Δ Moisture-weighted irrigated share — LUH2 $\times$ RECON [1960](#) [2015](#) [LD map](#)
- Why irrigation? 90% of consumptive water use ([Carleton et al., 2024b, 2025](#))

Instrument:

- Bartik IV: GAEZ irrigated suitability $\times$ commodity prices [maps](#) [prices](#)
- Alternative: yield-based IV

Weighting: optional cropland weighting (mean cropland share 1960–2015)

Controls: Δ GS potential evapotranspiration, Δ GS temperature — ERA5

Fact 2: data construction and modelling choices [back](#)

(a) Long difference: 1960→2015 [details](#)

- Early window: avg. 1958–1962; late window: avg. 2013–2017
- LUH2 anchor years at endpoints; GPCC/ERA5 smoothed over 5 years
- Captures long-run structural change, not year-to-year noise

(b) Growing-season precipitation [details](#)

- NH: Apr–Sep; SH: Jan–Mar + Oct–Dec
- Reduces noise from off-season months (sensitivity = annual precipitations)

(c) Gauge filter: GS gauge ≥ 1 [details](#) [gauge map](#)

- ≥ 1 active GPCC gauge in growing-season months, both windows
- Ensures measured (not interpolated) precipitation at endpoints

[LD precipitation](#)

[LD irrigation](#)

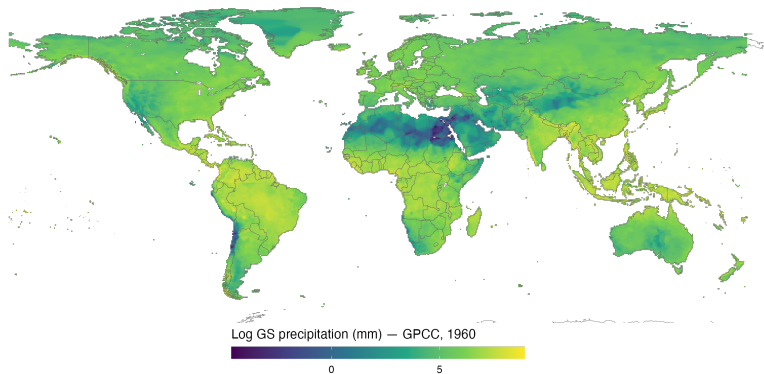
[LD exposure](#)

[LD cropland](#)

GS precipitation – GPCC, 1960

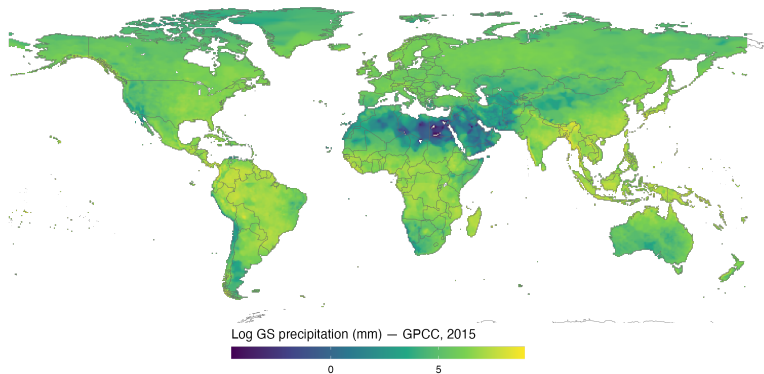
[back](#)

GPCC gauge-based growing-season precipitation at 0.5° . NH: Apr–Sep, SH: Oct–Mar.
Log-transformed for display.



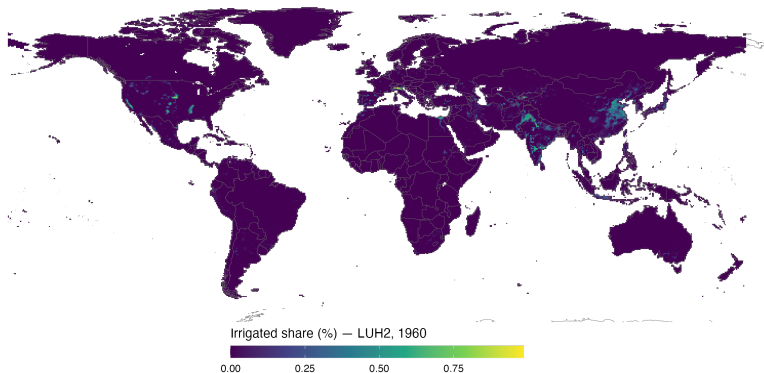
GS precipitation – GPCC, 2015 [back](#)

GPCC gauge-based growing-season precipitation at 0.5° . NH: Apr–Sep, SH: Oct–Mar.
Log-transformed for display.



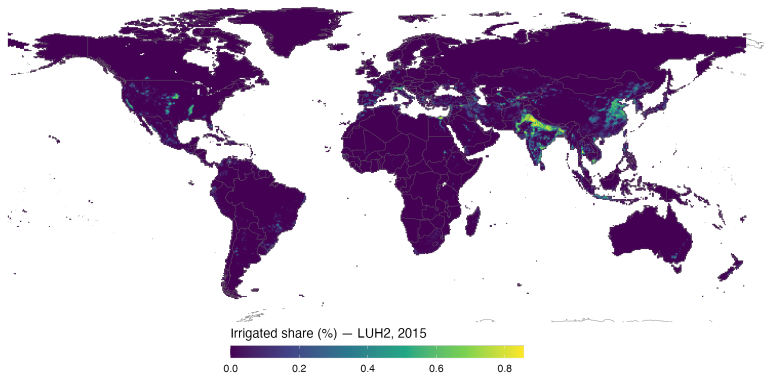
Irrigated land – LUH2, 1960 [back](#)

LUH2: share of grid cell under irrigation. Native 0.25° , resampled to 0.5° via area-weighted averaging.



Irrigated land – LUH2, 2015 [back](#)

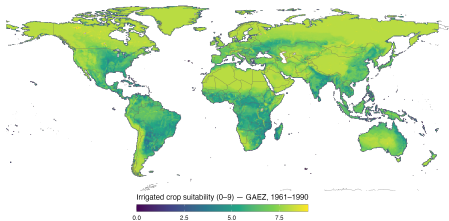
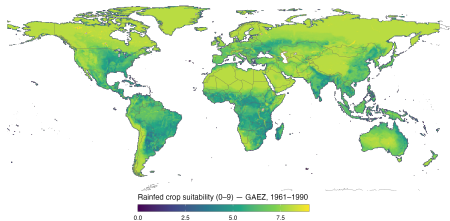
LUH2: share of grid cell under irrigation. Native 0.25° , resampled to 0.5° via area-weighted averaging.



Instrument for Δ irrigation share

- Bartik IV: GAEZ irrigated suitability index $\times \Delta$ commodity prices
- Alternative: yield-based (rather than suit. index) share $\times \Delta$ commodity prices

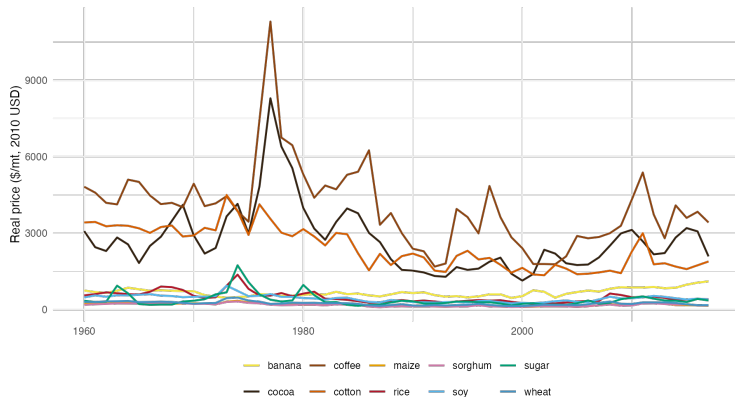
Figure: GAEZ v4 suitability index (0–9) for 10 crops, low-input, 1961–1990 climate baseline. Averaged across crops. 5-arcmin native, resampled to 0.5° . Left: rainfed suit. (Z_i^{crop}). Right: irrigated suit. (Z_i^{irrig}).



Bartik IV – commodity prices (World Bank)

[back](#)[back IV](#)

Annual real prices (\$/mt, 2010 USD) for 10 crops (wheat, rice, maize, sugar, cotton, soybean, sorghum, banana, coffee, cocoa). Used as time-varying shifters in Bartik IV: $\text{suitability}_i \times \text{price}_t$.

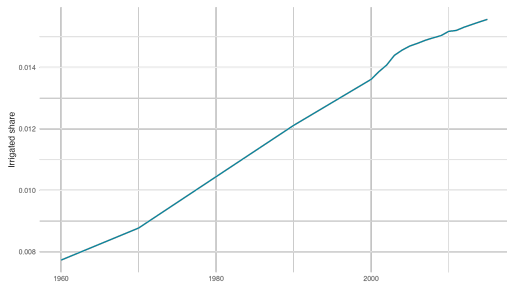


Long difference: 1960→2015 [back](#)

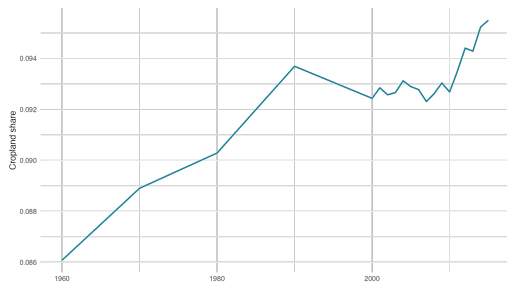
Long difference between endpoint averages: early (1958–1962) vs. late (2013–2017).

Data coverage:

- LUH2 irrigated/cropland shares: anchor years 1960 and 2015
- GPCC & ERA5: monthly, averaged over 5-year windows at each endpoint
- GAEZ crop suitability: 1961–1990 baseline (time-invariant)
- RECON atmospheric moisture: 2008–2017 average (single cross-section)



Irrigated share (LUH2)



Cropland share (LUH2)

GPCC growing-season precipitation

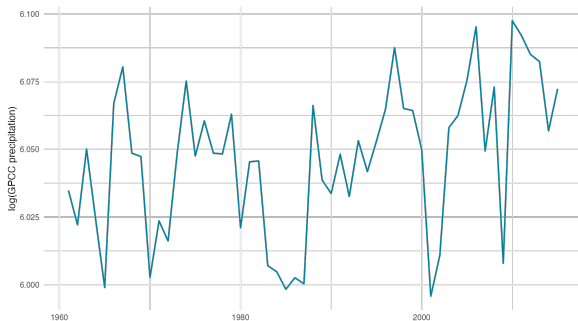
[back](#)[gauge map](#)

GPCC: monthly gauge-based precipitation, summed over growing-season months.

Growing season by hemisphere: NH = Apr–Sep, SH = Jan–Mar + Oct–Dec.

Advantages over ERA5 precipitation:

- Station-based measurement: **independent** of atmospheric reanalysis
- Avoids **mechanical correlation** with ERA5-derived controls (PET, temperature)
- GS gauge density for quality filtering (matched to outcome season)



Fact 2: sample selection

[back](#)[gauge map](#)

GPCC gauge count

- Each 0.5° cell $\times$ month: number of rain gauge stations contributing to the precipitation estimate
- Cells with 0 gauges rely entirely on interpolation from distant stations

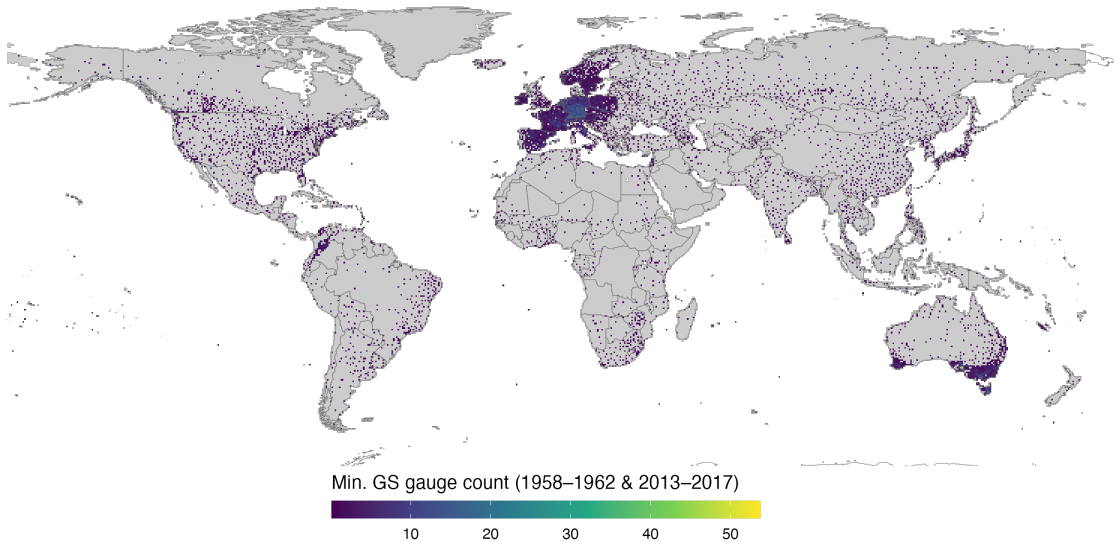
Growing-season gauge

- Mean gauge count over growing-season months only (NH: Apr–Sep, SH: Oct–Mar)
- Computed separately for each year, then take min across all years in both windows (1958–62, 2013–17)

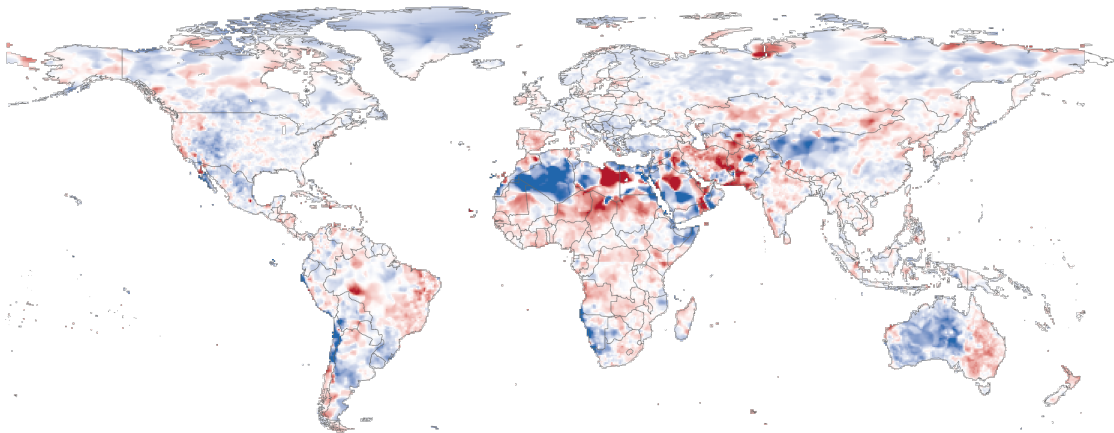
Filter: min GS gauge ≥ 1

- At least one active gauge in every year of both endpoint windows
- Same subsample used for both growing-season and annual precipitation outcomes

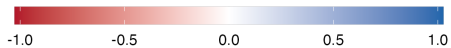
GPCC GS gauge density

[back main](#)[back](#)[sample](#)

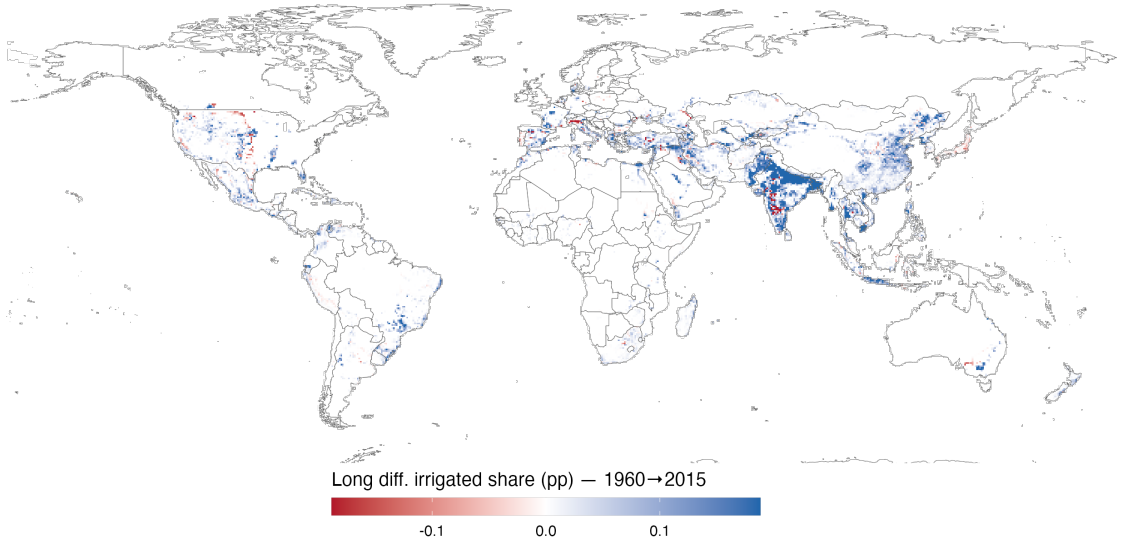
Long diff: GS precipitation

[back](#)[back main](#)

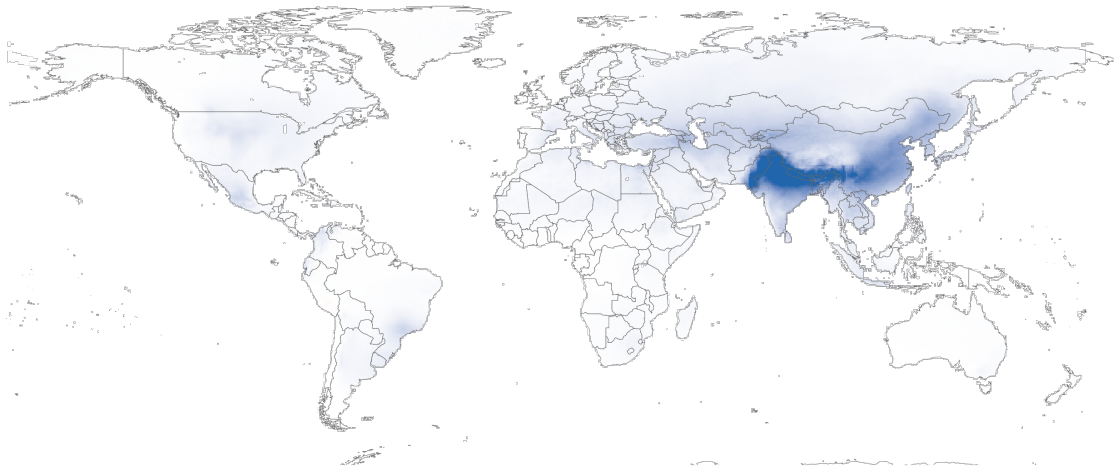
Long diff. log precip. (GS) — 1960→2015



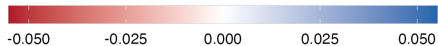
Long diff: irrigated share

[back](#)[back main](#)

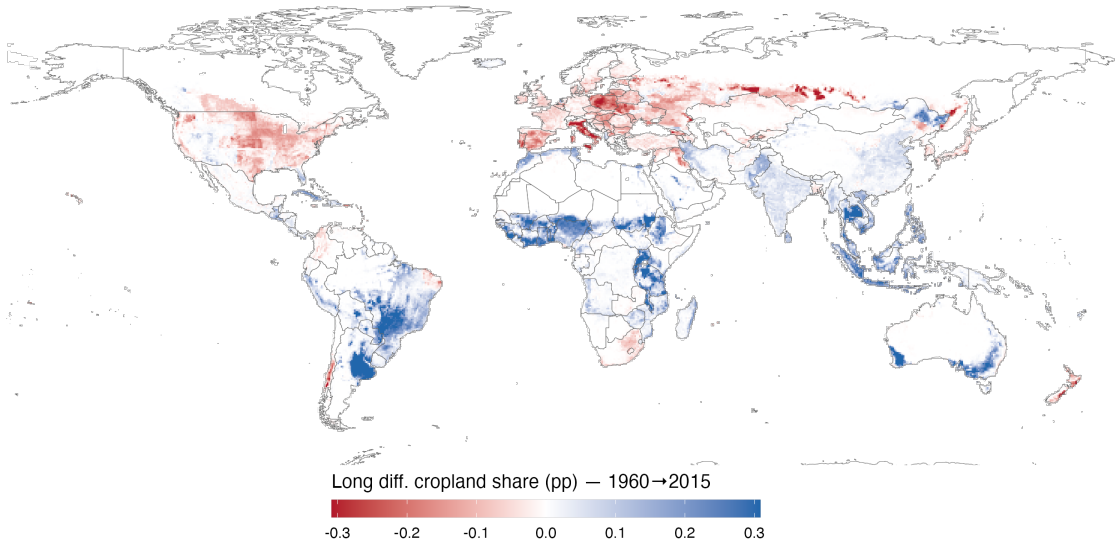
Long diff: irrigation exposure

[back](#)[back main](#)

Long diff. irrigation exposure — 1960→2015



Long diff: cropland share

[back](#)[back main](#)

Summary statistics: GS gauge ≥ 1 [back](#)

Variable	Mean	SD	P10	P50	P90
Growing-season precipitation					
Log GS precipitation, 1960	5.9286	0.8676	4.9781	5.9615	6.9080
Log GS precipitation, 2015	5.9346	0.8505	4.9201	5.9793	6.8916
Δ log GS precipitation	0.0060	0.2080	-0.2192	0.0100	0.2162
Annual precipitation					
Log annual precipitation, 1960	6.5587	0.7300	5.7625	6.5874	7.3629
Log annual precipitation, 2015	6.5765	0.7048	5.7972	6.5957	7.3699
Δ log annual precipitation	0.0178	0.1689	-0.1667	0.0183	0.1952
Irrigation exposure					
Irrigation exposure, 1960	0.0070	0.0106	0.0004	0.0037	0.0163
Irrigation exposure, 2015	0.0130	0.0192	0.0011	0.0070	0.0286
Δ irrigation exposure	0.0059	0.0097	0.0005	0.0030	0.0145
Irrigation intensity					
Irrigation intensity, 1960	0.0219	0.0306	0.0020	0.0119	0.0580
Irrigation intensity, 2015	0.0449	0.0536	0.0043	0.0274	0.1107
Δ irrigation intensity	0.0230	0.0273	0.0016	0.0147	0.0601
Controls and instruments					
Cropland share (mean 1960–2015)	0.2223	0.2124	0.0019	0.1575	0.5606
Δ GS PET (mm)	24.3627	41.0805	-18.9750	19.4425	80.4304
Δ GS temperature ($^{\circ}\text{C}$)	1.0414	0.5780	0.3860	0.9908	1.7912
Δ instrument (suitability)	-8549.0686	5134.9448	-15711.3110	-7472.4960	-2632.5599
Δ instrument (yield)	-494694.1071	463550.4846	-1099870.1190	-330605.2544	-108535.1101
Mean GS gauge count	2.2116	2.8127	1.0000	1.0000	5.0000

First stage: GS gauge ≥ 1 [back](#)

Dependent Variable:	Irrigated share			
Instrument	Suitability		Yield	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Suitability IV	-0.2110*** (0.0440)	-0.3088*** (0.0599)		
Yield IV			-0.5581*** (0.0865)	-0.5599*** (0.0975)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031
R ²	0.81440	0.89366	0.84250	0.89813

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

GS gauge ≥ 1 , GS precip: no climate controls [back](#)

Dependent Variable:	$\Delta \log$ precipitation (growing season)			
Estimator	OLS		IV	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Irrigated share	0.0422** (0.0197)	0.0409* (0.0217)	0.2280*** (0.0778)	0.2169** (0.0842)
Climate controls	No	No	No	No
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031
Kleibergen-Paap			20.257	29.863

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

GS gauge ≥ 1 , GS precip: yield IV [back](#)

Dependent Variable:	Δ log precipitation (growing season)			
Estimator	OLS		IV	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Irrigated share	0.0331*	0.0353*	0.0743**	0.1239**
	(0.0181)	(0.0198)	(0.0302)	(0.0524)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031
Kleibergen-Paap			40.672	36.146

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Intensive margin: irrigation intensity [back](#)

Dependent Variable: Estimator Weights Model:	$\Delta \log$ precipitation (growing season)			
	OLS		IV	
	None (1)	Cropland (2)	None (3)	Cropland (4)
<i>Variables</i>				
Irrigation intensity	0.0372** (0.0183)	0.0644** (0.0253)	0.1317*** (0.0403)	0.1610*** (0.0430)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031
Kleibergen-Paap			61.313	83.784

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Endogenous variable: Δ RECON-weighted irrigation intensity (irrigated share / cropland share). Same suitability IV.

GS gauge ≥ 1 , annual precip

[back](#)
[no climate](#)
[yield IV](#)

Dependent Variable: Estimator Weights Model:	$\Delta \log$ precipitation			
	OLS		IV	
	None (1)	Cropland (2)	None (3)	Cropland (4)
<i>Variables</i>				
Irrigated share	0.0370*** (0.0135)	0.0447** (0.0176)	0.2006*** (0.0642)	0.2930*** (0.0716)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031
Kleibergen-Paap			23.551	32.441

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

GS gauge ≥ 1 , annual: no climate [back](#)

Dependent Variable: Estimator Weights Model:	$\Delta \log \text{precipitation}$			
	OLS		IV	
	None (1)	Cropland (2)	None (3)	Cropland (4)
<i>Variables</i>				
Irrigated share	0.0403*** (0.0150)	0.0516*** (0.0196)	0.2250*** (0.0705)	0.2959*** (0.0720)
Climate controls	No	No	No	No
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031
Kleibergen-Paap			20.257	29.863

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

GS gauge ≥ 1 , annual: yield IV [back](#)

Dependent Variable:	$\Delta \log \text{precipitation}$			
Estimator	OLS		IV	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Irrigated share	0.0370*** (0.0135)	0.0447** (0.0176)	0.0636** (0.0268)	0.1262** (0.0522)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031
Kleibergen-Paap			40.227	35.043

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

All cells, GS precip

[back](#)[no climate](#)[yield IV](#)[annual](#)

Dependent Variable: Estimator Weights Model:	Δ log precipitation (growing season)			
	OLS		IV	
	None (1)	Cropland (2)	None (3)	Cropland (4)
<i>Variables</i>				
Irrigated share	-0.0057 (0.0163)	-0.0045 (0.0146)	0.3097 (0.3161)	0.2069*** (0.0718)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	67,630	42,576	67,630	42,576
Kleibergen-Paap			2.2350	33.141

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

All cells, GS: no climate [back](#)

Dependent Variable:	$\Delta \log$ precipitation (growing season)			
Estimator	OLS		IV	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Irrigated share	0.0042 (0.0175)	0.0062 (0.0154)	0.3665 (0.3531)	0.2045*** (0.0779)
Climate controls	No	No	No	No
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	67,630	42,576	67,630	42,576
Kleibergen-Paap			2.1956	31.614

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

All cells, GS: yield IV [back](#)

Dependent Variable:	$\Delta \log$ precipitation (growing season)			
Estimator	OLS		IV	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Irrigated share	-0.0057 (0.0163)	-0.0045 (0.0146)	-0.0364 (0.0640)	0.0266 (0.0401)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	67,630	42,576	67,630	42,576
Kleibergen-Paap			25.995	37.743

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

All cells, annual precip

[back](#)[no climate](#)[yield IV](#)

Dependent Variable: Estimator Weights Model:	$\Delta \log$ precipitation			
	OLS		IV	
	None (1)	Cropland (2)	None (3)	Cropland (4)
<i>Variables</i>				
Irrigated share	-0.0047 (0.0134)	-0.0036 (0.0113)	0.1495 (0.2396)	0.2590*** (0.0789)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	67,630	42,576	67,630	42,576
Kleibergen-Paap			2.1866	32.478

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

All cells, annual: no climate [back](#)

Dependent Variable:	$\Delta \log \text{precipitation}$			
Estimator	OLS		IV	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Irrigated share	0.0051 (0.0134)	0.0010 (0.0112)	0.1613 (0.2392)	0.2578*** (0.0787)
Climate controls	No	No	No	No
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	67,630	42,576	67,630	42,576
Kleibergen-Paap			2.1956	31.614

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

All cells, annual: yield IV [back](#)

Dependent Variable:	$\Delta \log \text{precipitation}$			
Estimator	OLS		IV	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Irrigated share	-0.0047 (0.0134)	-0.0036 (0.0113)	-0.0861* (0.0514)	0.0233 (0.0372)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	67,630	42,576	67,630	42,576
Kleibergen-Paap			24.772	35.367
Conley (500km) standard-errors in parentheses				
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1				

Imported vs. domestic irrigation exposure [back](#)

Dependent Variable: Estimator Weights Model:	Δ log precipitation (growing season)			
	OLS		IV	
	None	Cropland	None	Cropland
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Imported irrigated share	0.0301* (0.0175)	0.0318 (0.0196)	0.2090*** (0.0721)	0.2500*** (0.0768)
Domestic irrigated share	0.0055 (0.0037)	0.0038 (0.0042)	0.0489** (0.0222)	0.0079 (0.0089)
Climate controls	Yes	Yes	Yes	Yes
Cragg-Donald F	—	—	259.7	293.7
1st stage F : imported	—	—	14.0	16.2
1st stage F : domestic	—	—	12.4	16.8
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Endogenous variables: Δ RECON-weighted imported and domestic irrigated shares (in SD). Separate Bartik IVs for each component. Kleibergen-Paap F not computable (near-collinear projected instruments under Conley SEs); table reports Cragg-Donald F (iid) and per-equation 1st-stage Wald F (Conley).

Placebo: downwind irrigation exposure [back](#)

Dependent Variable:	Δ log precipitation (growing season)			
Estimator	OLS		IV	
Weights	None	Cropland	None	Cropland
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Irrigated share (downwind, resid.)	-0.0444*** (0.0137)	-0.0369** (0.0171)	-0.0124 (0.0581)	0.0280 (0.0649)
Climate controls	Yes	Yes	Yes	Yes
<i>Fixed-effects</i>				
Region-Aridity quintile-Latitude band	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,367	5,031	5,367	5,031
Kleibergen-Paap			18.421	21.435

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Downwind = transposed RECON matrix (moisture sent to, not received from). Residualized on upwind exposure to remove the shared spatial component ($\rho = 0.73$). IV first stage KP $F > 24$.

Horse race: upwind vs. downwind (OLS) [back](#)

Dependent Variable:	Δ log precipitation (growing season)	
Estimator	OLS	
Weights	None	Cropland
Model:	(1)	(2)
<i>Variables</i>		
Irrigated share	0.0640*** (0.0219)	0.0594** (0.0280)
Irrigated share (downwind)	-0.0714*** (0.0230)	-0.0573** (0.0286)
Climate controls	Yes	Yes
<i>Fixed-effects</i>		
Region-Aridity quintile-Latitude band	Yes	Yes
<i>Fit statistics</i>		
Observations	5,367	5,031

Conley (500km) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Both upwind and downwind irrigation exposure (in SD) enter jointly.

General equilibrium [back](#)

Static equilibrium – wages $\{w_i\}$ simultaneously clear:

$$\text{(Goods)} \quad Q_{i,k} = \sum_n \bar{d}_{in,k} C_{in,k} \quad \text{(Water)} \quad \frac{L_{i,0}}{G_i(R_i)} = \sum_{k=1}^K H_{i,k} \quad \text{(Labor)} \quad \bar{L}_i - L_{i,0} = \sum_{k=1}^K L_{i,k}$$

Dynamic equilibrium: sequence of static equilibria $\{w_i(t)\}_t$ with $R_i(0)$ given

- Stocks $\{R_i(t)\}_t$ evolving per the LOM above (state variable)

Steady-state equilibrium: dynamic equilibrium satisfying the $R_B^{(ss)}$ fixed-point

[solving algor.](#)

[simulations](#)

Solving algorithm: static equilibrium [back](#)

Iterative algorithm for static GE given water stocks $\{R_i\}$ and tax rates $\{\tau_{i,k}\}$:

1. **Initialize:** guess $\{w_i, I_i\}$; normalize wages to numeraire w_{i_0}
2. **Prices:** compute $r_i = w_i G_i(R_i)$; unit costs $c_{i,k}$; cost shares $\alpha_{i,k}$; producer and consumer prices $P_{ji,k}, \tilde{P}_{ji,k}$
3. **Quantities:** compute $C_{ji,k}$; revenues $Y_{i,k}$; water use $H_{i,k} = (1 - \alpha_{i,k})Y_{i,k} / (r_i + \tau_{i,k})$; water labor $L_{i,0} = G_i(R_i) \sum_k H_{i,k}$; tax revenues T_i
4. **Update & check:** $w_i^{(new)} = \frac{\sum_k \alpha_{i,k} Y_{i,k}}{\bar{L}_i - L_{i,0}}$ (Labor MCC); $I_i^{(new)} = w_i \bar{L}_i + T_i$

→ Stop if $\max_i |(w_i^{(new)} - w_i) / w_i|$ and $\max_i |(I_i^{(new)} - I_i) / I_i|$ sufficiently small;
else dampen and return to Step 1

Solving algorithm: dynamic & steady state [back](#)

Dynamic equilibrium: solve static GE each period t ; update stocks via the LOM:

$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \mathbf{\Phi} \mathbf{R}_B(t) + \mathbf{\Psi}(t) \mathbf{H}(t)$$

Steady-state equilibrium: fixed-point intersection of two mappings

- (a) Static GE $\rightarrow \mathbf{H}$ as a function of $\mathbf{R}_B^{(ss)}$
- (b) LOM fixed point: $\mathbf{R}_B^{(ss)} = (\mathbf{I} - \mathbf{\Phi})^{-1} (\mathbf{\Gamma} + \mathbf{\Psi}^{(ss)} \mathbf{H}(\mathbf{R}_B^{(ss)}))$

$\Rightarrow$ Iterate on $\mathbf{R}_B^{(ss)}$ until both mappings are mutually consistent

Optimal water policy – water stock linearization [back](#)

Re-write steady-state water stocks (from the LOM fixed point) as:

$$\mathbf{R}_B^{(ss)} = \mathbf{Z} + \mathbf{\Lambda}^{(ss)} \mathbf{H}^{(ss)}, \quad \mathbf{Z} \equiv (\mathbf{I} - \mathbf{\Phi})^{-1} \mathbf{\Gamma}, \quad \mathbf{\Lambda}^{(ss)} \equiv (\mathbf{I} - \mathbf{\Phi})^{-1} (\mathbf{F}_{BB}^{(E)} - \mathbf{I}) \boldsymbol{\theta}(t)$$

$$R_i^{(ss)} = Z_i + \sum_j \sum_k \lambda_{ji,k} H_{j,k}^{(ss)} \quad [\text{location level}]$$

i.e., $\lambda_{ji,k} \equiv$ hydrological "impact factor" of withdrawals at (j, k) on steady-state R_i

Optimal policy: dual problem [back](#)

Dual reformulation: planner directly chooses consumer prices

$$\mathbb{P}_i \equiv \{\tilde{P}_{ii,k}, \tilde{P}_{in,k}, \tilde{P}_{ni,k}; \tilde{r}_{i,k}\}_{n \neq i,k}$$

Income function for country i :

$$I_i(\cdot) = w_i L_i + \underbrace{(\tilde{r}_i - r_i)^\top H_i}_{\text{water taxes}} + \underbrace{(\tilde{P}_i - P_i)^\top C_i}_{\text{tariffs + prod. taxes}} + \underbrace{(\tilde{P}_{-i} - P_{-i})^\top C_{-i}}_{\text{export taxes}}$$

Key functions in the dual:

- Producer price: $P_{ij,k}(w_i, \tilde{r}_{i,k}) = \bar{d}_{ij,k} c_{i,k}(w_i, \tilde{r}_{i,k})$
- Water cost: $r_i(w_i, R_i) = w_i G_i(R_i)$
- Steady-state stock: $R_i = Z_i + \sum_{j,k} \lambda_{ji,k} H_{j,k}$

Optimal policy: wage neutrality [back](#)

Lemma 1 (Local wage neutrality): $\frac{\partial I_i(\cdot)}{\partial w_i} = 0$

Proof: expand using Shephard's lemma + labor market clearing:

$$\bar{L}_i - \underbrace{G_i(R_i) \sum_k H_{i,k}}_{=L_{i,0}} - \underbrace{\sum_k \frac{\alpha_{i,k} P_{ii,k} Q_{i,k}}{w_i}}_{=\sum_k L_{i,k}} = 0 \quad \checkmark$$

Wages only redistribute b/w wage payments and tax revenues when the planner controls all prices.

Assumption A1: policy-induced changes in *foreign* relative wages have no first-order effect on home income at the optimum.

Corollary 1: $\frac{\partial I_i(\cdot)}{\partial \mathbf{w}} = \mathbf{0} \Rightarrow$ wages drop from all FOCs

Optimal policy: first-order conditions [back](#)

FOC w.r.t. any policy instrument $\tilde{P}$ (after applying Corollary 1):

$$\sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial \tilde{P}} + \sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} \left[\frac{\partial I_n}{\partial \tilde{P}} + \frac{\partial I_n}{\partial R} \frac{\partial R}{\partial \tilde{P}} + \frac{\partial I_n}{\partial H} \frac{\partial H}{\partial \tilde{P}} + \frac{\partial I_n}{\partial C} \frac{\partial C}{\partial \tilde{P}} \right] = 0$$

Key derivatives (1/2):

- **Direct price effect (Roy's id.):** $\frac{\partial V_n}{\partial \tilde{P}} = -\frac{\partial V_n}{\partial E_n} C_{in,k}$ if $\tilde{P} \in \tilde{P}_n$, else 0
- **Direct income, water:** $\frac{\partial I}{\partial \tilde{r}_{i,k}} = 0$ (water tax revenue and cost changes cancel exactly)

Optimal policy: key derivatives (cont.)

[back](#)

Key derivatives (2/2):

- **Income via water stocks:** $\frac{\partial I_i}{\partial R_i} = -\frac{G'_i(\cdot)}{G_i(\cdot)} r_i \sum_k H_{i,k}$
- **Income via water use:** $\frac{\partial I_i}{\partial H_{i,k}} = \tilde{r}_{i,k} - r_i$
- **Policy impact on stocks:** $\frac{\partial R_i}{\partial \tilde{P}} = \sum_{j,k} \lambda_{ji,k} \frac{\partial H_{j,k}}{\partial \tilde{P}}$

Optimal policy: decomposition of FOCs [back](#)

FOCs decompose into three independent sub-problems:

$$\left\{ \begin{array}{ll} \sum_n \left(\frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} \pi_n \right) = \frac{\partial \mathcal{W}}{\partial W_i} \frac{\partial V_i}{\partial E_i} & \text{(distributional objective)} \\ \tilde{\mathbf{P}} = \mathbf{P} & \text{(efficient non-water prices)} \\ \tilde{r}_{i,k} - r_i = \sum_n \frac{G'_n}{G_n} r_n \sum_k H_{n,k} \sum_g \lambda_{in,g} & \text{(efficient water prices)} \end{array} \right.$$

Complex GE derivatives $\partial \mathbf{C} / \partial \tilde{\mathbf{P}}$, $\partial \mathbf{H} / \partial \tilde{\mathbf{P}}$ cancel in both the goods and water FOCs.

Optimal policy: solutions [back](#)

Solutions to the three sub-problems:

- $\pi_i^* = \frac{(\partial \mathcal{W} / \partial W_i)(\partial V_i / \partial E_i) I_i}{\sum_n (\partial \mathcal{W} / \partial W_n)(\partial V_n / \partial E_n) I_n} \Rightarrow \Delta_i^* = \pi_i^* \sum_n I_n - I_i$
- $t_{ij,k} = x_{ij,k} = s_{i,k} = 0$
- $\tau_{j,g} = \sum_i \frac{G'_i(\cdot)}{G_i(\cdot)} r_i \sum_k \lambda_{ji,g} H_{i,k}$

Optimal policy – international transfers [back](#)

Optimal income shares achieving any distributional objective:

$$\pi_i^* = \frac{\frac{\partial \mathcal{W}}{\partial W_i} \frac{\partial V_i}{\partial E_i} I_i}{\sum_n \frac{\partial \mathcal{W}}{\partial W_n} \frac{\partial V_n}{\partial E_n} I_n}, \quad \Delta_i^* = \pi_i^* \sum_n I_n - I_i$$

Special case: log SWF $\mathcal{W} = \sum_i \omega_i \ln W_i$ + homothetic demand

- $\pi_i^* = \omega_i \Rightarrow$ efficient frontier traced by varying $\{\omega_i\}$

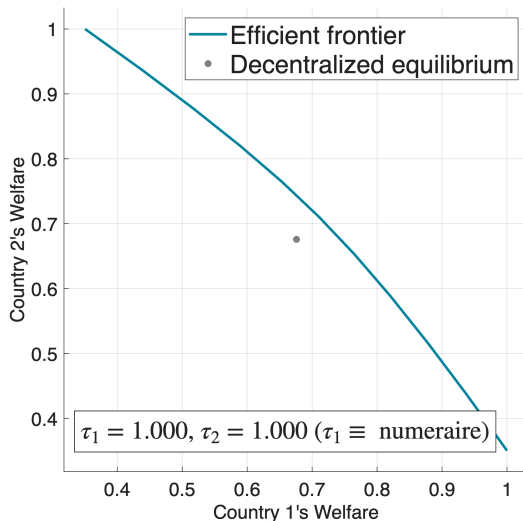
Optimal policy separates efficiency from distribution: water taxes correct the externality; transfers achieve the distributional goal

Optimal water policy – illustration

[back](#)[back quant.](#)

Two symmetric countries + ocean

- $f_{ii}^{(E)} = 10\%$, $f_{ij}^{(E)} = 50\%$, $f_{i0}^{(E)} = 40\%$
- Equally productive in two identical $k > 0$ sectors (final goods)



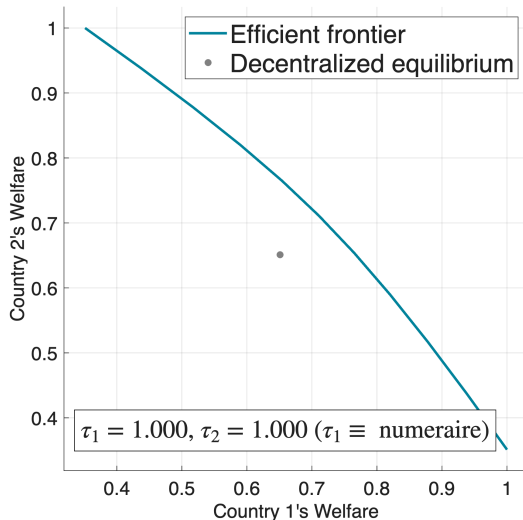
Optimal water policy – illustration

[back](#)[back quant.](#)

Two symmetric countries + ocean

- $f_{ii}^{(E)} = 10\%$, $f_{ij}^{(E)} = 50\%$, $f_{i0}^{(E)} = 40\%$
- Equally productive in two identical $k > 0$ sectors (final goods)

More inefficient if $f_{ij}^{(E)} = 5\%$ & $\uparrow \theta_{i,k}$ & G'

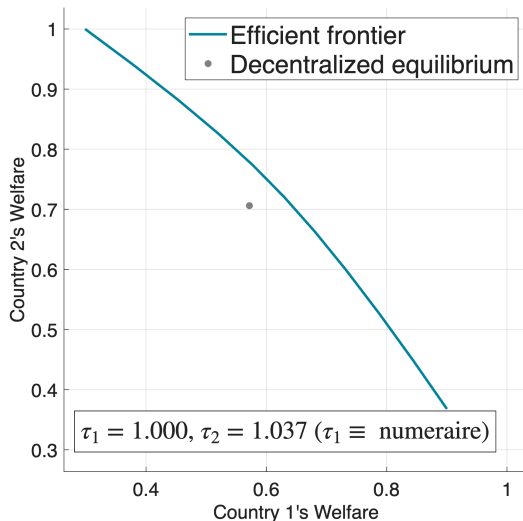


Optimal water policy – illustration

[back](#)[back quant.](#)

Atmospheric asymmetry $f_{12}^{(E)} > f_{21}^{(E)}$

– $\lambda_{21} > \lambda_{12} \rightarrow$ **higher** τ_2



Optimal water policy – illustration

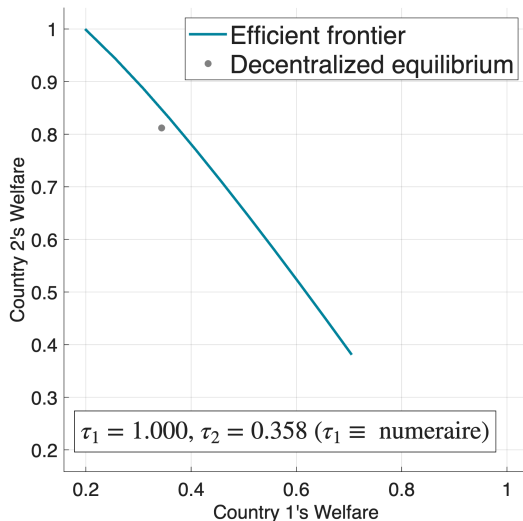
[back](#)[back quant.](#)

Atmospheric asymmetry $f_{12}^{(E)} > f_{21}^{(E)}$

- $\lambda_{21} > \lambda_{12} \rightarrow$ **higher** τ_2

Higher τ_1 if economically asymmetric

- Country 1 productive in $k = 1$ (agric.)
- Country 2 prod. in $k = 2$ (non-agric.)
- That is, $\theta_1 > \theta_2 \rightarrow \lambda_{12} > \lambda_{21}$



Quantification – water module [back](#)

Our method quantifies $\{\kappa_i, \delta_i, \Gamma_i, f_{ij}^{(E)}, f_{ij}^{(S)}\}_{i,j \in \mathcal{B}}$ for **any geography set** $\mathcal{B}$. Starting with

$$\mathbf{R}_B(t+1) = \mathbf{\Gamma} + \left(\mathbf{I} - \boldsymbol{\kappa} - \boldsymbol{\delta} + \mathbf{F}_{BB}^{(E)} \boldsymbol{\kappa} + \mathbf{F}_{BB}^{(S)} \boldsymbol{\delta} \right) \mathbf{R}_B(t)$$

1. Measure $R_i(t)$, $E_i(t)$ and $S_i(t)$ for a long-run, "humanless" scenario (1901-20)
2. Measure $f_{ji}^{(E)}$ and $f_{ji}^{(S)}$ directly from data
3. Calculate $\kappa_i = E_i(t) / R_i(t)$ and $\delta_i = S_i(t) / R_i(t)$
4. Assuming $\Delta R_i(t) = 0$, calculate $\mathbf{\Gamma} = \left(\boldsymbol{\kappa} + \boldsymbol{\delta} - \mathbf{F}_{BB}^E \boldsymbol{\kappa} - \mathbf{F}_{BB}^S \boldsymbol{\delta} \right) \mathbf{R}_B(t)$

[hydrology models](#)[atmosph. flows](#)[surface flows](#)[κ_i, δ_i, Γ_i: results](#)[validation](#)

Step 1: water data from hydrology models [back](#)

Table 5: Water Models for Total Water Storage (TWS), Evaporation, and Surface Runoff Data

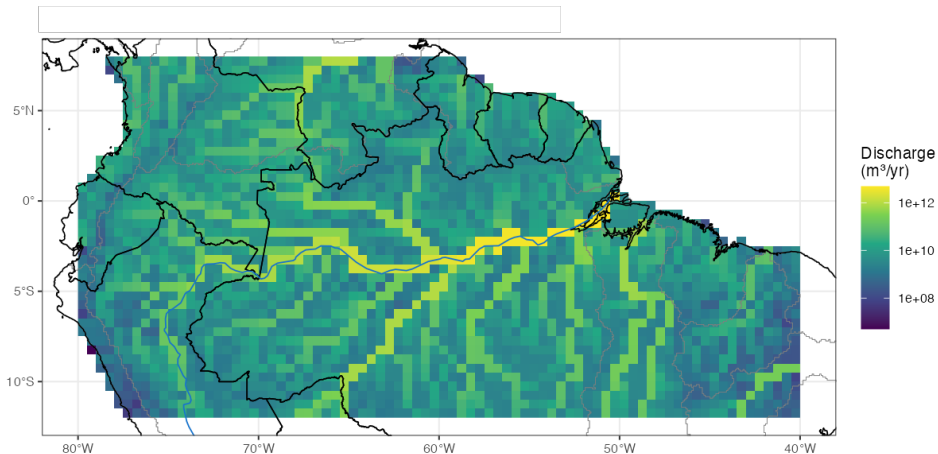
Model	Model Info	ISIMIP	Time Coverage/ Scenario	Meteo	TWS Meas.	Mean TWS (km ³)	GW	DGW	TWS components: ¹⁴					
									Soil	Riv	Lk	Snw	Can	
clm45	Community Land Model 4.5 ¹	2b	pre-industrial	GCM	relative	—	x		x	x		x	x	
cwatm	Community Water Model v1.04 ²	2b	both	GCM	absolute	234,727	x		x	x	x	x	x	
cwatm	Community Water Model v1.06 ²	3a, 3b	historical	reanalysis	absolute	204,787	x		x	x	x	x	x	
ho8	Ho8 ³	2b	both	GCM	absolute	57,042	x		x	x	x	x		
ho8	Ho8, newer calibration ³	3a, 3b	historical	reanalysis	unspecified	46,748	x		x	x	x	x		
hydropy	Newer MPI-HM ⁴	3a	historical	reanalysis	absolute	54,993	x		x	x	x	x	x	
jules-es-55	Joint UK Land Env. Simulator v5.5 ⁵	2b	historical	GCM	unspecified	595,826	x		x		x			
jules-es-vn6p3	Joint UK Land Env. Simulator v6.3 ⁵	3a, 3b	historical	reanalysis	unspecified	2,233,363			x					
jules-w2	Joint UK Land Env. Simulator v6.2 ⁵	3a, 3b	historical	reanalysis	absolute	113,572			x	x		x	x	
lpjml	LPJ managed Land ⁶	2b	both	GCM	unspecified	147,074			x	x	x	x		
lpjml5-7-10-fire	LPJ managed Land 5.7.10-fire ⁶	3a	historical	reanalysis	absolute	138,317			x	x	x	x		
matsiro	MIROC-INTEG + land integration ⁷	2b	both	GCM	absolute	6,094,733	x	x	x	x	x	x	x	
miroc-integ-land	MATSIRO + land integration ⁸	3a, 3b	historical	reanalysis	absolute	6,073,546	x	x	x	x	x	x	x	
mpi-hm	Older HydroPy ⁴	2b	both	GCM	unspecified	98,313	x		x	x		x		
orchidee-mict	Orchidee ⁹	3a	historical	reanalysis	unspecified	124,279			x			x		
pcr-globwb	PCRaster Global Water Balance ¹⁰	2b	both	GCM	both	153,787	x	x	x	x	x	x	x	
watergap2-2c	WaterGAP2-2c ¹¹	2a, 2b	both	GCM	relative	—	x		x	x	x	x	x	
watergap2-2e	WaterGAP2-2e ¹²	3a, 3b	historical	reanalysis	relative	—	x		x	x	x	x	x	
web-dhm-sg	Water and Energy Budget-Based ¹³	3a, 3b	historical	reanalysis	unspecified	97,838			x	x				

Notes: ¹ Oleson et al. (2013) ² Burek et al. (2020) ³ Hanasaki et al. (2008); Hanasaki and Yamamoto (2010) ⁴ Stacke and Hagemann (2021) ⁵ Best et al. (2011); Wiltshire et al. (2020) ⁶ Schaphoff et al. (2018) ⁷ Pokhrel et al. (2017) ⁸ Yokohata et al. (2020) ⁹ Risi et al. (2016) ¹⁰ Sutanudjaja et al. (2018) ¹¹ Müller Schmied et al. (2021) ¹² Müller Schmied et al. (2023) ¹³ Gosling et al. (2023) ¹⁴ GW stands for renewable/young ground-water storage (recharged from precipitation), DGW for deep groundwater (stored in deep and old, fossil-fuel-like reservoirs), Riv for rivers, Lk for lakes, Snw for snow, and Can for canopy water storage components.

Step 2: quantifying surface flows $f_{ji}^{(S)}$

[back](#)[back quant](#)

Logic: combine cumulative water runoff raster + directional flow raster

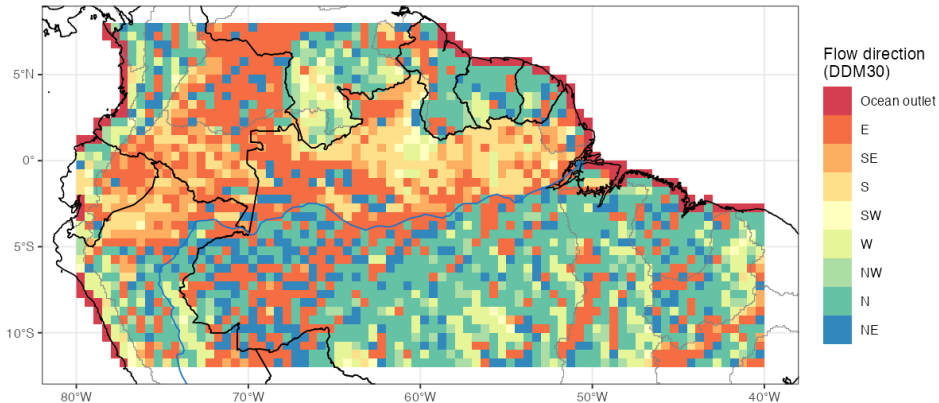


Step 2: quantifying surface flows $f_{ji}^{(S)}$

[back](#)[back quant](#)

Logic: combine cumulative water runoff raster + directional flow raster

DDM30 flow direction (colored) — Amazon basin

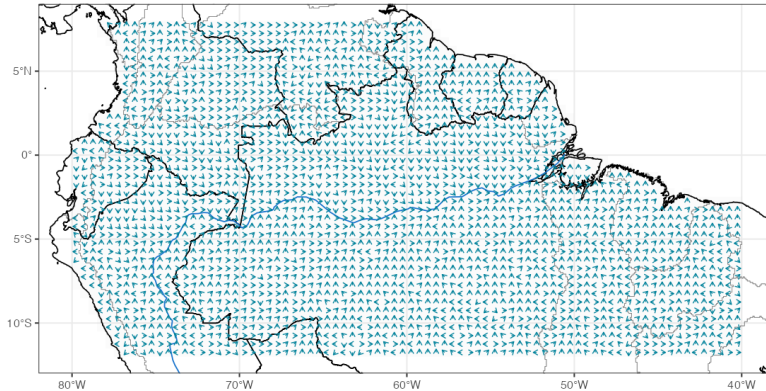


Step 2: quantifying surface flows $f_{ji}^{(S)}$

[back](#)[back quant](#)

Logic: combine cumulative water runoff raster + directional flow raster

DDM30 flow direction (arrows) — Amazon basin



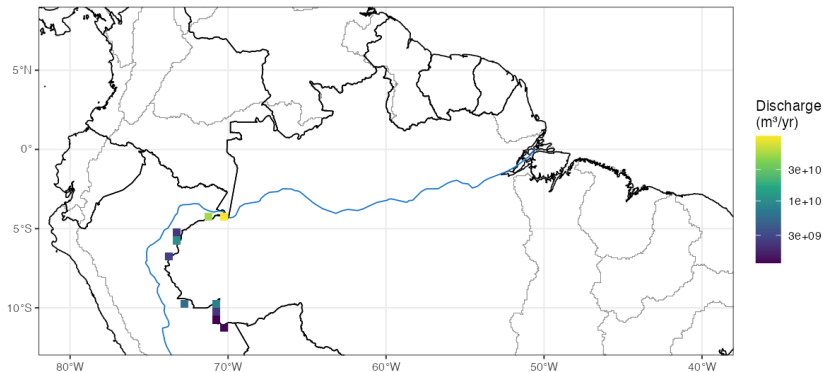
Step 2: quantifying surface flows $f_{ji}^{(S)}$

[back](#)[back quant](#)

Logic: combine cumulative water runoff raster + directional flow raster

Peru → Brazil cross-border discharge pixels (cwatm)

10 transition pixels



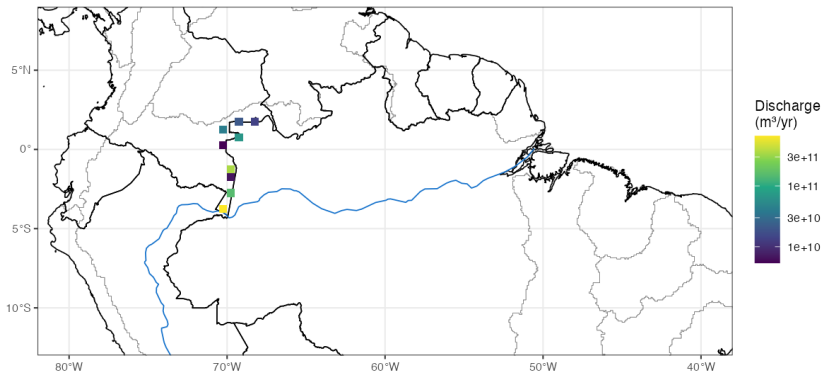
Step 2: quantifying surface flows $f_{ji}^{(S)}$

[back](#)[back quant](#)

Logic: combine cumulative water runoff raster + directional flow raster

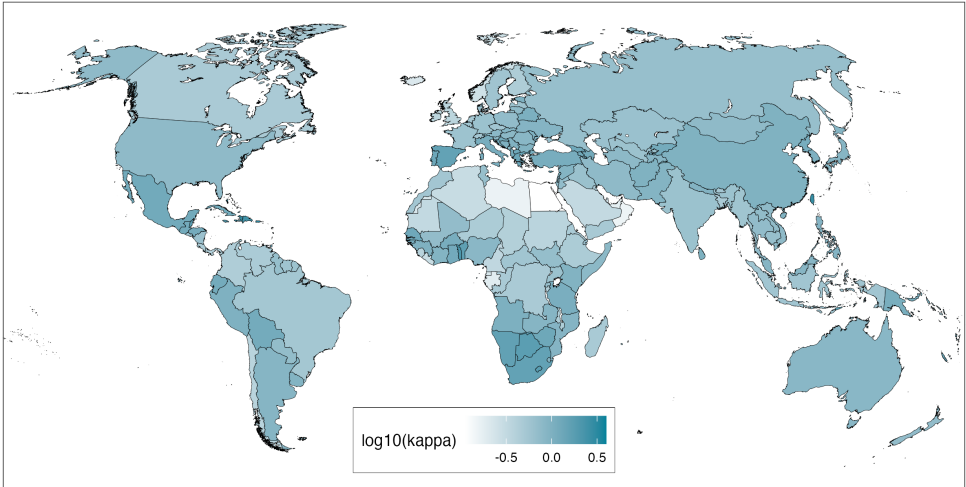
Colombia → Brazil cross-border discharge pixels (cwatm)

9 transition pixels



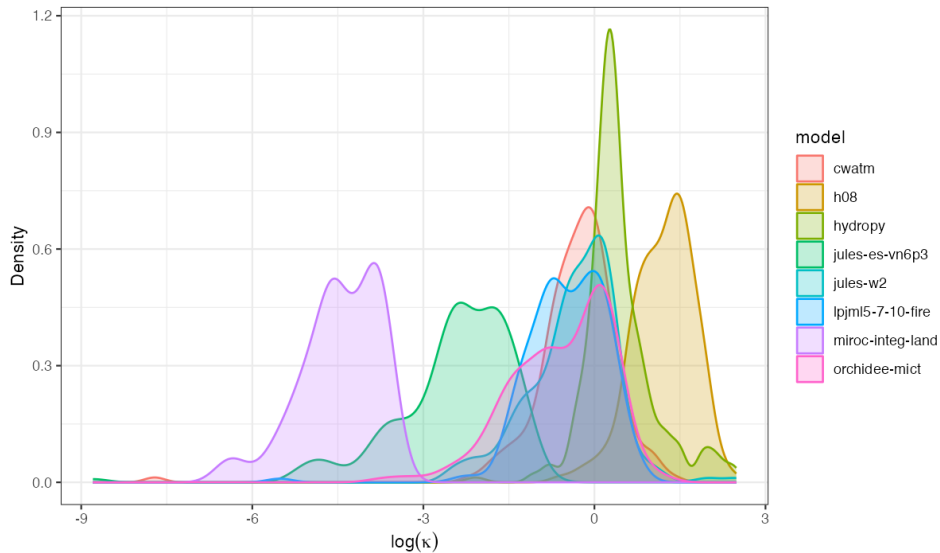
Step 3: quantified $\{\kappa_i, \delta_i, \Gamma_i\}$ [back](#)

Kappa (evap / stock) — cwatm



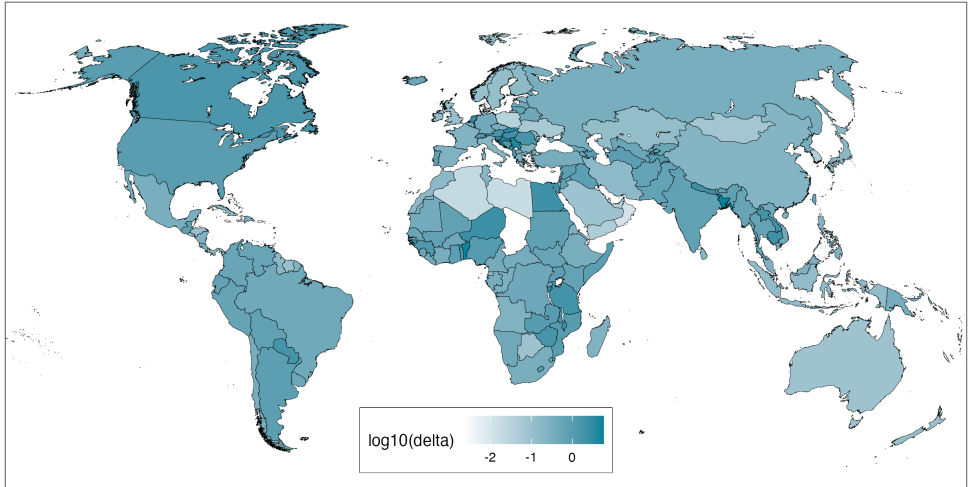
Step 3: quantified $\{\kappa_i, \delta_i, \Gamma_i\}$ [back](#)

Distribution of $\log(\kappa)$ across countries, by model



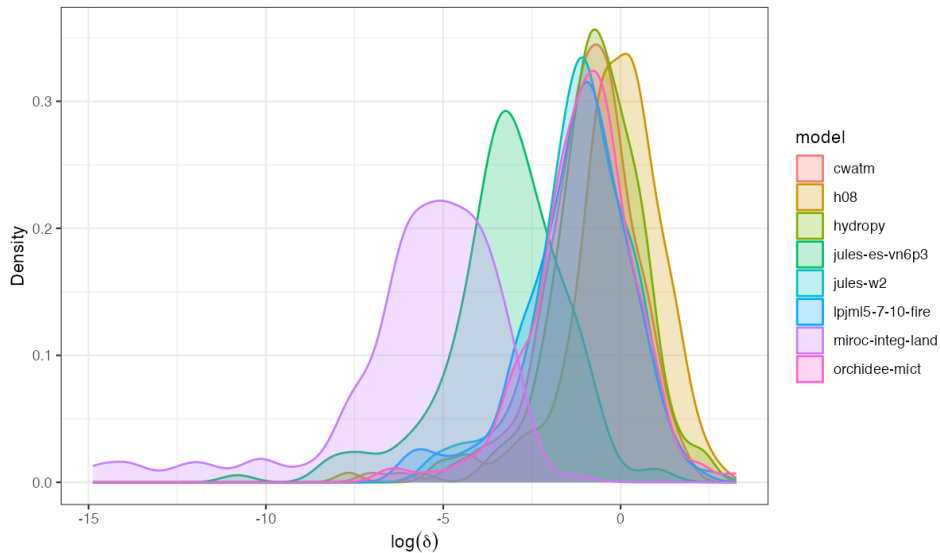
Step 3: quantified $\{\kappa_i, \delta_i, \Gamma_i\}$ [back](#)

Delta (runoff / stock) — cwatm



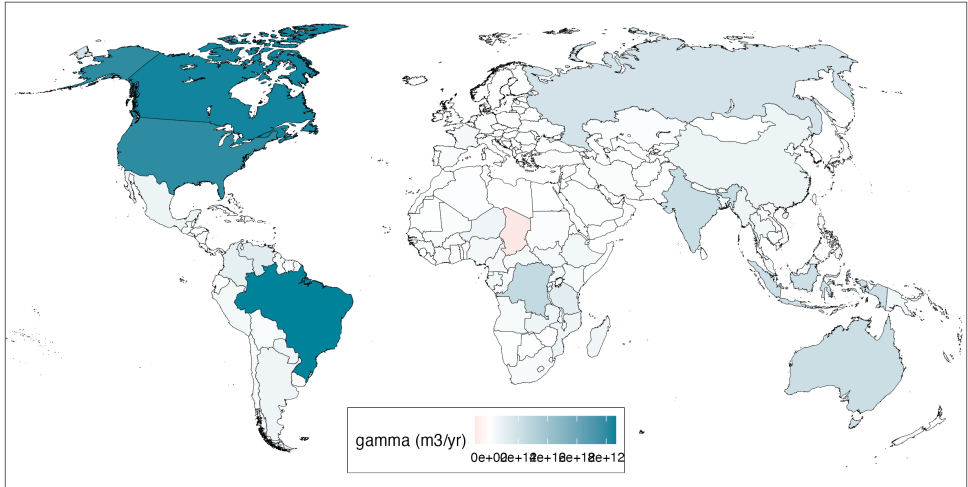
Step 3: quantified $\{\kappa_i, \delta_i, \Gamma_i\}$ [back](#)

Distribution of $\log(\delta)$ across countries, by model (delta = runoff / stock)



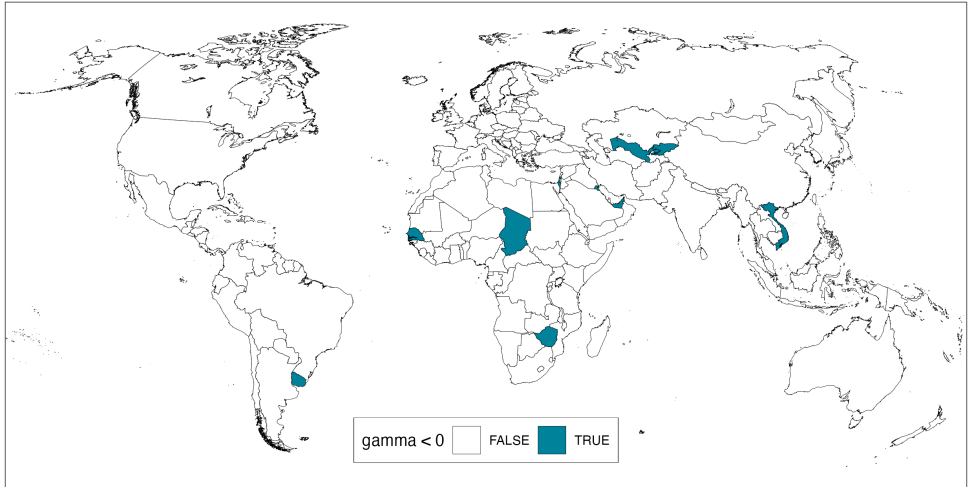
Step 3: quantified $\{\kappa_i, \delta_i, \Gamma_i\}$ [back](#)

Gamma (ocean precip needed for water balance) — cwatm



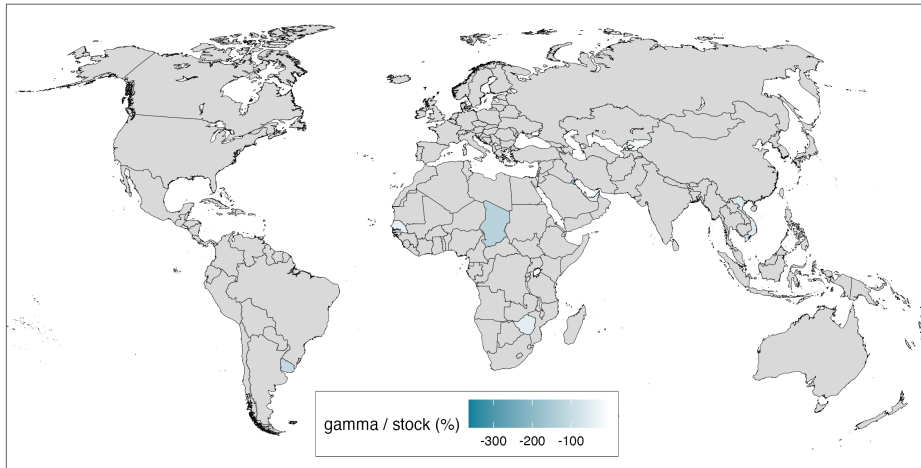
Step 3: quantified $\{\kappa_i, \delta_i, \Gamma_i\}$ [back](#)

Negative gamma (ocean precip needed < 0) — cwatm



Step 3: quantified $\{\kappa_i, \delta_i, \Gamma_i\}$ [back](#)

Gamma as % of stock (negative-gamma countries only) — cwatm



Grey = non-negative gamma (ratio not applicable).

Step 3: validating $\{\kappa_i, \Gamma_i\}$ [back](#)

Table 6: Validating estimated $\{\kappa_i\}_i$ (cwatm model)

Dependent Variable: Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				kappa				
<i>Variables</i>								
Forest cover share (broad)	0.0089 (0.1612)	0.0065 (0.1595)						
Desert/barren share			-0.6640*** (0.1368)	-0.6286*** (0.1383)				
GW / TWS ratio					-0.6893*** (0.1620)	-0.7038*** (0.1598)		
Avg. temperature (°C)							0.0098** (0.0049)	0.0071 (0.0051)
<i>Fixed-effects</i>								
Controls for stocks	No	Yes	No	Yes	No	Yes	No	Yes
<i>Fit statistics</i>								
R ²	1.72×10^{-5}	0.02674	0.11743	0.12903	0.09277	0.12331	0.02221	0.03717
Observations	179	179	179	179	179	179	179	179

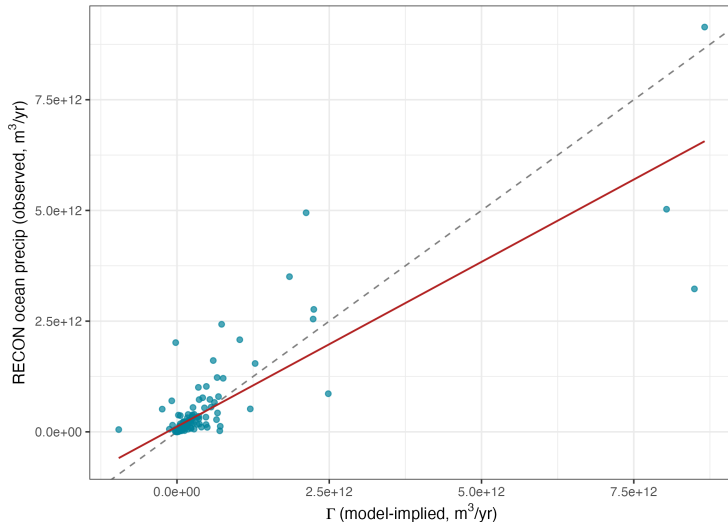
IID standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Cross-country OLS. Dependent variable: $\hat{\kappa}$ (evap / TWS, cwatm, 1901–1920). IID standard errors in parentheses. Broad forest = GAEZ values 2 & 6 (closed + needle/mixed forest).

Step 3: validating $\{\kappa_i, \Gamma_i\}$ [back](#)

Model-implied gamma vs. RECON ocean precipitation — cwatm



Spearman rho = 0.803 | Pearson R2 = 0.714

Calibrating θ_{ik} : consumptive-use shares [back](#)

$\theta_{ik} \in (0, 1)$: share of water withdrawn by sector k in location i that is consumed

Source: WaterGAP 2.2e gridded potential water use ([Müller Schmied et al., 2023](#))

- Sectoral withdrawal H_{ik}^{withd} and consumption H_{ik}^{cons} at 0.5° , monthly, 2000–2019
- Yearly totals $\rightarrow$ annual $\theta_{ik,t} = H_{ik,t}^{\text{cons}} / H_{ik,t}^{\text{withd}} \rightarrow$ average across years

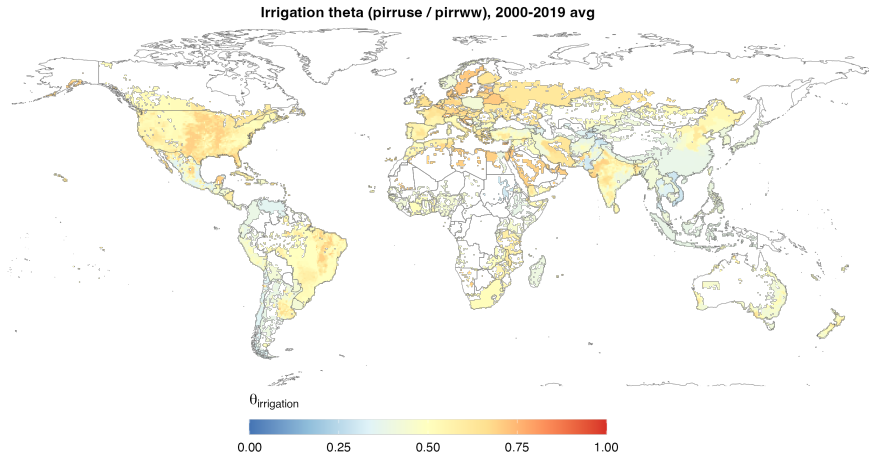
Sector	N cells	Mean	Median	SD	P10	P90	$\theta > 1$
Livestock	59399	1.000	1.000	0.000	1.000	1.000	0.000
Domestic	51762	0.192	0.151	0.138	0.102	0.252	0.000
Irrigation	27649	0.515	0.510	0.104	0.380	0.661	0.000
Manufacturing	12336	0.169	0.139	0.091	0.100	0.348	0.000
Thermoelectric	8248	0.193	0.007	0.348	0.005	0.712	0.007

[irrigation](#)[domestic](#)[thermoelectric](#)[manufacturing](#)[livestock](#)

θ_{ik} : irrigation [back](#)

Irrigation water lost to crop evapotranspiration. Varies with technology and local climate.

WaterGAP sub-model: Global Irrigation Model (GIM), truly gridded at 0.5° . Irrigated areas ([Siebert et al., 2015](#)); crop calendars ([Portmann et al., 2010](#)); irrigation efficiency by technology ([Jägermeyr et al., 2015](#)).

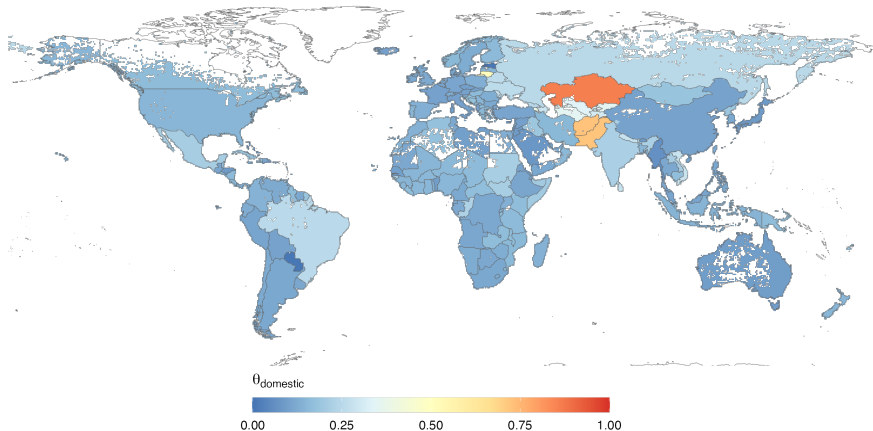


θ_{ik} : domestic [back](#)

Water consumed by households and small businesses (drinking, cooking, sanitation).

WaterGAP sub-model: structural country-level model ([Flörke et al., 2013](#)); driven by GDP p.c., population, and technological change. Downscaled to 0.5° by population density.

Domestic theta (pdomuse / pdomww), 2000-2019 avg

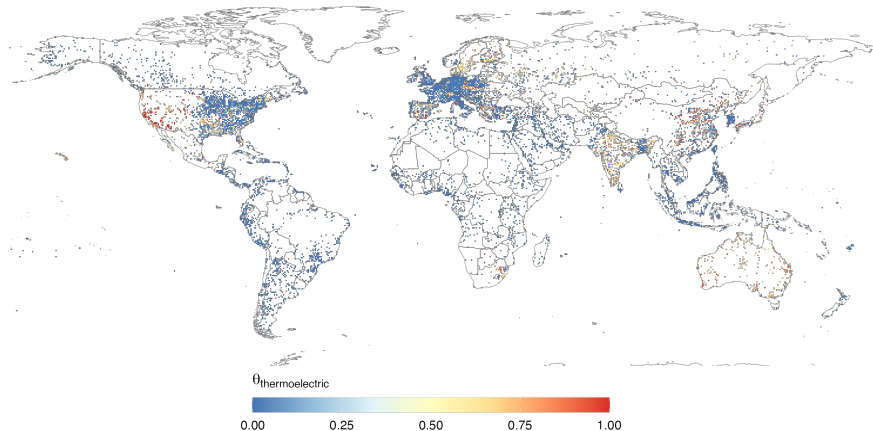


θ_{ik} : thermoelectric [back](#)

Cooling water for thermal power plants.

WaterGAP sub-model: cooling-technology-specific estimates ([Vassolo and Döll, 2005](#)); plant-level data aggregated to country, downscaled to 0.5° by thermal power plant locations.

Thermoelectric theta (pelecuse / pelecww), 2000-2019 avg

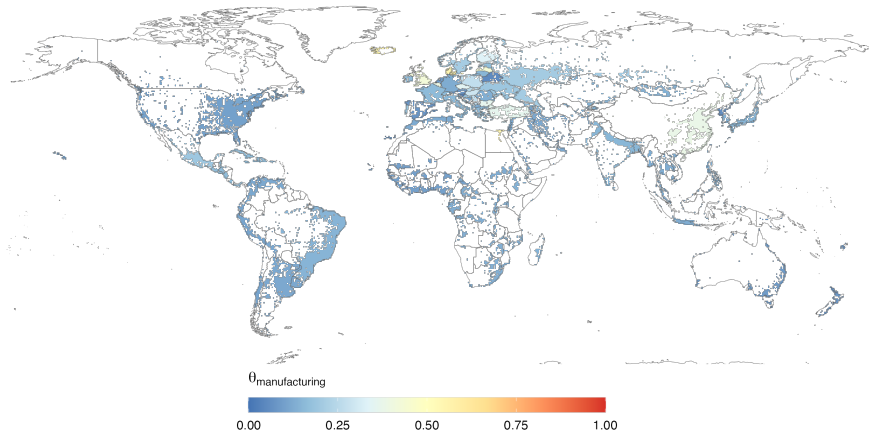


θ_{ik} : manufacturing [back](#)

Process and cooling water in industry.

WaterGAP sub-model: country-level manufacturing VA $\times$ water intensity (Flörke et al., 2013);
downscaled to 0.5° by urban population density.

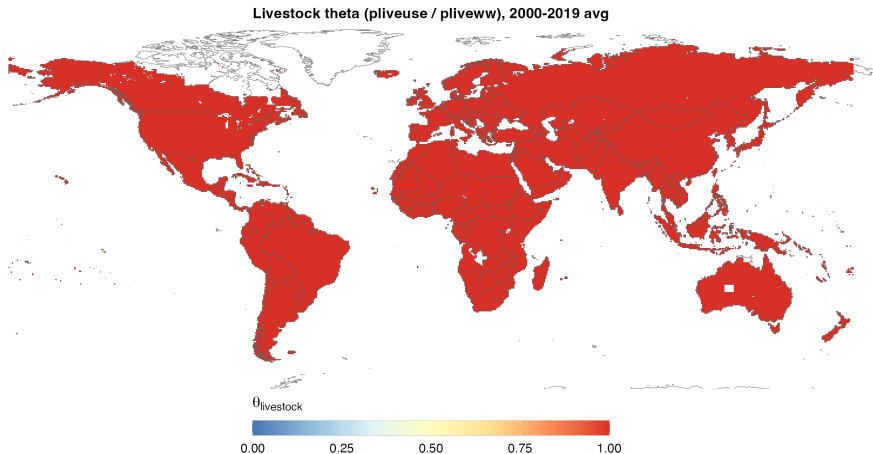
Manufacturing theta (pmanuse / pmanww), 2000-2019 avg



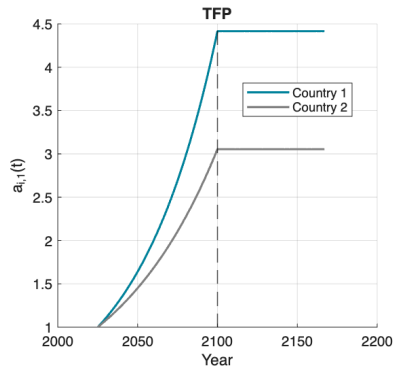
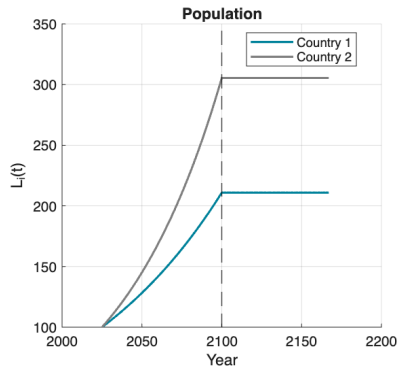
θ_{ik} : livestock [back](#)

Drinking water for animals. All water consumed (metabolized or transpired).

WaterGAP sub-model: per-head drinking water needs ([Alcamo et al., 2003](#)); gridded via FAO Gridded Livestock of the World maps.



Illustrational simulations

[back quant.](#)[back GE](#)

Illustrational simulations

[back quant.](#)[back GE](#)

2-Country Variant (ex1 derived): Time Paths (dashed = terminal SS, dotted = growth stops)

